Complex & Intelligent Systems (2025) 11:443
https://doi.org/10.1007/s40747-025-02055-9

ORIGINAL ARTICLE

Robot movement based inference of teleoperator state for multi-robot search

Arunim Bhattacharya[1] · **Sachit Butail**[1]

Received: 18 December 2024 / Accepted: 5 August 2025 / Published online: 8 September 2025
© The Author(s) 2025

Abstract

Human behavior encodes important information about how we perceive and act in a new environment. Extracting such information indirectly from behavioral observations has the potential to advance natural human–robot interaction without the need for wireless transmission of sensitive data. This is especially relevant in wilderness search and rescue missions that use unmanned aerial vehicles (UAVs), to cover large areas. In such missions, while human teleoperators typically have prior knowledge about the environment or the missing person, they also experience mental workload, and may not be situationally aware at all times. An effective human–robot team, where teleoperators are assisted by autonomous robots, should function so that the autonomous robots are able to adapt their search strategy in response to human state—knowledge, workload, and situational awareness—based on movement of the teleoperated UAV. Accordingly, in this paper we test the hypotheses that teleoperator knowledge and state can be estimated from movement cues and that such cues can then be used to adaptively steer autonomous robots toward an improved search performance. We conduct human-subject experiments in a virtual replica of a section of the Grand Canyon National Park, which reports one of the largest number of missing person cases every year. Participants were selectively briefed with prior knowledge about the local terrain and the missing person before every trial. EEG and eye tracking data were collected to directly measure cognitive load and situational awareness through multiple measures. Our results show that both search performance and turn rate of teleoperated vehicle depended on missing person knowledge. We further find that cognitive load and situational awareness as measured through physiological measures are weakly correlated with movement behavior. To use these results for adaptive search, we estimate prior knowledge and situational awareness from movement data and use it in a human-swarm robot interaction strategy where multiple robots adapt their separation and proximity to the human teleoperated UAV. Our simulation results show that adaptive search performs nearly as well as a comprehensive spiral based search in terms of success rate and is able to do so in less time. Results from this work set the stage for new non-verbal human robot interaction strategies in field settings.

Keywords Missing person search · Teleoperation · Human–multi-robot interaction · Prior knowledge · Situational awareness · Cognitive load

Introduction

Between 1992 and 2007, there have been over 65,000 search and rescue (SAR) incidents in the National Parks in United States [1]. In 2005 alone there were over 2400 SAR operations with the five national parks—Grand Canyon National Park, Gateway National Recreation Area, and Yosemite National Park, Lake Mead National Recreation Area, and Rocky Mountain National Park—accounting for close to fifty percent of such incidents, and the Grand Canyon National park in Arizona reporting the most such operations. Without an effective response, one in five such incidents would result in a fatality [1].

In a wilderness search and rescue (WiSAR) operation, one of the first steps is to establish a point last seen (PLS) or last known point (LKP), depending on where the missing person was last sighted or assumed present. Once a PLS is established, search areas are drawn by analyzing lost person behavior and terrain. The search areas are then prioritized

✉ Sachit Butail
sbutail@niu.edu

Arunim Bhattacharya
Z1776960@students.niu.edu

[1] Department of Mechanical Engineering, Northern Illinois University, 590 Garden Rd., Dekalb, IL 60115, USA

based on metrics such as probability of area (POA), which is the probability that the lost person is contained within a particular region [2]. Because the search area grows geometrically with time, locating a lost person alive is a time-critical task.

Conventional WiSAR operations involve teams of searchers along with rescue animals on the ground coordinating with central operations to comb a large area spanning tens of square kilometers [3]. During such operations, searchers and rescuers often use small aircrafts and helicopters equipped with thermal cameras to sweep inaccessible locations from air [3]. Indeed, over the past decade, unmanned aerial vehicles (UAV) are being increasingly used in WiSAR missions to find missing persons [4–6]. While completely autonomous solutions that enable terrain following [7], and swarm based solutions that cover large areas [8] are part of ongoing research, a common use of UAVs in WiSAR involves human teleoperation with a live camera feed [9, 10].

A WiSAR mission with teleoperated UAVs involves skilled operators who are expected to search efficiently within a region that has a high probability of finding the missing person. In doing so, the operators are presumed to take into account missing person behavior and terrain features [11, 12]. In reality, however, while teleoperators may be skilled in flying a UAV, they may not be familiar with the terrain they are flying over [13], or fully understand the behavior profile of the missing person in terms of how fast they may be going, their mental state, or whether they will stay near the trails [14]. Trails may form temporarily and represent a path of least resistance for the missing person to walk on [15]. While this information about the missing person and trails may be available to the teleoperators through other members of the WiSAR team, retaining and recovering this knowledge while flying the UAV can impose cognitive burden. Thus, prior knowledge, not skill, can be an important factor in search performance. Indeed, as has been shown in missing person models, such prior knowledge about the missing person and the terrain can significantly affect mission performance by improving the accuracy of search regions [16, 17].

In this context, inferring prior knowledge from UAV movement in the field can inform rapidly responsive control strategies. For example, in a human-swarm interaction (HSI) setup where the operator interacts directly with a robot swarm on the field by teleoperating a single UAV [18], such inference can be used to indirectly inform other autonomous robots (also UAVs but referred to as robots to distinguish the role) in the field, who may then utilize adaptive control strategies to selectively search near or far from the teleoperated UAV [19, 20].

Because teleoperators have to assimilate a large amount of information in a very little time, they are also bound to experience varying degrees of cognitive burden [21, 22], which in turn may affect their performance. Relatedly, situational awareness (SA) is another key factor that can influence operator performance [23, 24] and is typically measured by querying operators on relevant information or through physiological measures such as EEG or eye-tracking [25]. Inferring cognitive load and situational awareness directly from the movement of the UAV that is controlled by the teleoperator, therefore, has the benefit of fine-tuning the control response of autonomous robots nearby without having to wirelessly transmit sensitive EEG and eye tracking data. Indeed, such capabilities have the potential to expand the use of nonverbal human–robot interaction in the field from gesture based communication directly from the operator to indirect observation of the instinctual movements of the vehicle they operate [26].

Accordingly, this work is motivated by the following hypotheses. For a teleoperator searching for a missing person, the movement of the teleoperated UAV is affected by: (H1) prior knowledge of terrain or missing person, (H2) cognitive load of a teleoperator, and (H3) situational awareness of the teleoperator. To test these hypotheses, we conduct an experimental study where participants instrumented with an EEG and an eye-tracking headset search for a missing person in a virtual replica of a 10×10 km section of the Grand Canyon National Park. To artificially create prior knowledge, we selectively reveal aspects of the environment and the missing person to participants before they begin the search. To additionally investigate if the presence of other UAVs affect teleoperator search behavior we place the robots either clustered in a random location or spread uniformly through the environment. Teleoperator movement is quantified in terms of speed, turn rate, and time spent still in place (freezing), and time spent turning in place (turning while still). Our results show that while prior knowledge affects speed and turn rate, cognitive load as measured using EEG and eye-tracking data is weakly correlated to speed, turn rate, and freezing; situational awareness measured using dwell time and α power is also weakly correlated to the movement behaviors.

The experimental results are next used to test the hypothesis (H4) that an HSI strategy where autonomous robots respond to prior knowledge and SA of the human teleoperator in real time will perform better than one that doesn't. In particular, a long short-term memory (LSTM) based classifier is designed to directly infer prior knowledge from movement data of the teleoperated UAV and a linear regression is used to estimate normalized α power of the EEG signal which serves as an indirect estimate of SA. The observation window for the LSTM classifier is selected to optimize classification accuracy. These classifiers are then integrated into new HSI control strategies that regulate autonomous robot assistance to the human based on their inferred prior knowledge and SA. These strategies are compared to other approaches where

multiple robots search independently along a random path or search in a spiral pattern.

The contributions of this work are as follows: (i) we test the hypotheses that movement features of a teleoperated UAV are dependent on prior knowledge of the missing person and terrain, cognitive load, and situational awareness; (ii) we evaluate machine learning (ML) methods to classify prior knowledge and situational awareness from movement cues to identify the observation window that maximizes the accuracy; and (iii) we validate human-swarm interaction strategies that adaptively regulate swarm assistance in search based on inference of teleoperator prior knowledge and situational awareness. The motivation and approach in this work matches a recent study where a Bayesian classifier was used to infer prior knowledge and validate an information-theoretic control strategy for adaptive search in a laboratory environment [27]. However, there are important distinctions in terms of (a) the search region that is orders of magnitude larger here and uses a teleoperated UAV, (b) the type of prior knowledge, which is different owing the type of search mission and more representative of missing person search in a large park, (c) measuring and integrating EEG and eye tracking data into testing new hypotheses for inferring and using cognitive load and situational awareness in search, and (d) the adoption and update of commonly used swarm control algorithms to assist a human teleoperator during search.

This paper is organized as follows: "Related work" reviews related work on studies that investigate teleoperator assistance strategies in SAR through intelligent interface design, dynamically changing POAs based on input from ground operations, and shared autonomy approaches where teleoperator input is blended with autonomous actions. "Background" describes relevant background information on prior knowledge, cognitive load, and situational awareness (SA), including how cognitive load and SA can be determined using EEG and pupillometry data. "Problem statement" states the WiSAR problem with HSI context. "Experimental study" presents the experimental study including methods and results related to hypotheses H1, H2, and H3. This is followed by "Estimating human prior knowledge and state based on movement cues" which describes how we estimate prior knowledge with an LSTM and SA with linear regression. "Adaptive search based on prior knowledge and SA" describes the simulation study including the multi-robot control strategies used to assist the human teleoperator simulation setup and results, which compare the proposed search with other random and spiral search strategies in terms of success rate and the time taken to find the missing target. We discuss the experimental and simulation results in "Discussion" followed by a conclusion in "Conclusion and future directions".

Related work

Teleoperating a single UAV is a common mode for search and rescue operations [28]. While purely autonomous solutions are increasingly proposed in the literature, the level of risk associated with missing person search makes human involvement for detection and rescue inevitable. A WiSAR mission that utilizes teleoperated UAVs is expected to involve skilled teleoperators who may not be familiar with the local terrain [29]. Furthermore, information about the missing person themselves may be interpreted and integrated differently into search strategies by each teleoperator [13]. Interviews with SAR responders in regions with extreme climates regarding the use of UAVs have highlighted the need to assimilate vast amounts of additional information including environmental conditions and legal restrictions in addition to dealing with local capacities [30].

In this context, assistance to the teleoperator may be provided in the form of interface design which enables improved search regions [31–33], reduced cognitive load [34], and greater situational awareness [23, 34, 35]. Assistance may also be provided in the form of coordinated efforts between ground and aerial searchers [11], in the form of shared autonomy [36], where an intelligent UAV may selectively ignore or blend human control input with its own inference of the mission, or in the form of multiple autonomous robots that may intelligently search the region to minimize search time [37]. This work is motivated by a scenario where autonomous robots may assist a human teleoperator by inferring and acting on their prior knowledge, cognitive load, and SA.

Accessing human knowledge and skill towards improved efficiency in search is not new. For example, critical information from UAV teleoperators can be integrated with lost person behavior simulations to take search decisions [31, 32]. Niedzielski et al. demonstrate the effectiveness of a coordinated search and rescue strategy where teleoperators search POAs as they are assisted by imagery analysts to identify the missing person, who is then rescued by ground searchers [11]. The POAs on the map are drawn based on a lost-person mobility model that takes into account the terrain features to delineate areas of easy access by the missing person. More recently, long-short-term memory (LSTM) networks have been used to predict human actions that are integrated into a game theoretic model to act as a decision aid to plan, search, and compute lost person trajectories [33], and transfer learning has been used predict human rescuers' navigation and triage strategies in urban search and rescue (USAR) missions [38].

Shared autonomy approaches include blended autonomy, where a mediation agent blends command from the teleoperator with the autonomous actions of robots based on sensory input [36], adaptive autonomy [37] where the autonomous robots decide themselves which tasks to exe-

cute, adjustable autonomy [39, 40], where the tele-operator assigns tasks to the robots, and mixed initiative [37, 41], where the robots and teleoperator switch between tasks depending on the level of success. Assistance from a single robot based on intent recognition was studied in [42] where a UAV acts on first-responder's movements to search within a simulated urban environment. Assistance from autonomous robots in a multi-robot setup have been explored in mutual information blending strategy [19] to provide a context independent search strategy for coordination between a teleoperator and autonomous robots. This study focuses on enabling an adaptive search through the inference of prior knowledge, cognitive load, and situational awareness from non-verbal movement cues. The use of movement cues allows communication-free and privacy-preserving implementation of robotic swarm assistance in the field.

Background

Prior knowledge

Prior knowledge refers to relevant information about the task at hand that could be retrieved to improve task efficiency. Prior knowledge is different from skill; for example, one may consider a situation where a skillful UAV operator may be searching an area not known to them. Prior knowledge has been shown to require additional cognitive capacity [43].

Examples of prior knowledge about the lost person include mental state, fatigue, and physical state, and has been shown to reduce search times significantly if considered in agent-based models [44]. Prior knowledge about terrain can include elevation, vegetation, and trails, and may be used to guide search actions such as searching within a particular region or near trails [45]. While such information can be provided to a skilled UAV operator prior to a mission, it may not be effectively utilized at all times. Furthermore, in the absence of a clear mental model of the teleoperator, it is difficult to quantify whether prior knowledge is properly utilized by a teleoperator in their search [13].

Cognitive load

Cognitive load is the mental workload experienced by a human as they perform a task [46]. During a complex time intensive-task such as search and rescue, cognitive load can play a critical role in modulating task performance [47]. Specifically task performance may suffer when teleoperators have to make decisions under time pressure and cognitive limits are exceeded due to excessive information load or memory recall requirements [48].

Depending on the application, cognitive load can be measured through survey responses [49], secondary task performance [50], electroencephalogram (EEG) data [51–53], and eye-tracking data [52]. In this work, to robustify the measurement of cognitive load, we adopt a multi-pronged approach whereby we measure cognitive load using EEG and eye-tracking data. Specifically for EEG, based on prior work, we utilize the data from electrodes that measure brain activity from the frontal lobe to calculate weighted difference in spectral power over a fixed duration [50, 53]. Additionally, we use eye tracking data to calculate saccades and pupil dilation which have been shown as indicators of cognitive load [54–56].

Situational awareness

Situational awareness (SA) [57] refers to the ability to perceive, comprehend, and anticipate elements in the environment that are critical for decision-making and effective action. It involves three levels: paying attention to the environment, interpreting it correctly, and predicting how events will unfold [57].

Being situationally aware can improve safety, decision-making, and performance across a wide range of activities and time-intensive missions [58]. The conventional approach of measuring SA is by freezing the simulation (freeze-framing) to pose a Situation Awareness Global Assessment Technique (SAGAT) questionnaire that pose contextually relevant questions designed to test the three different levels of SA. However, posing such questions in the middle of a mission distract participants from the goal at hand and interrupt with truly measuring movement based correlates of SA. Therefore, in this study we infer the first level of situational awareness which is generally associated with visual attention [59]. Specifically, we measure SA from eye tracking data in terms of dwell time within an area of interest [60–63].

Problem statement

Consider a WiSAR scenario where a team comprising a single human teleoperated UAV robot and a swarm of autonomous robots are searching for a missing person. Due to the large search area, the human teleoperator can only see the autonomous robot positions in an overhead view available in an inset on their feed. Compared to the human teleoperator, the autonomous robots have a lower missing person detection rate.

We assume no communication between the teleoperator and the autonomous robots other than that they are each aware of each other's pose in real time. This is because information about knowledge or cognitive state may create security vulnerabilities [64], and require additional bandwidth; information from the autonomous robots to the teleoperator such as video streams may pose additional cognitive burden.

The robot swarm moves according to a swarming model [65] which enables collision avoidance and coordination in motion as the robots stay together. Each robot can further regulate its motion according to inference about the teleoperator drawn based on their pose data. The goal of the mission is to find a missing person as soon as possible.

Experimental study

The experimental study was designed to test hypotheses H1-H3 and comprised participants teleoperating a UAV through a large virtual environment to look for a missing person as their prior knowledge is varied. We additionally varied the arrangement of a swarm of quasi-static robots within the region to evaluate the effect of swarm robot distribution on search behavior.

Virtual environment

The virtual environment consisted of a 10 km × 10 km replica of a section of the Grand Canyon National Park (Fig. 1). The park terrain was built in Unity software using height-map data obtained from an online heightmap generator [66]. The location of the terrain was approximately centered at coordinates 36°5′57″ N and 112°5′38″ W. The terrain was further enriched using trees and static water marking the Colorado River. The light brown color of the landscape was selected to match pictures of the park.

A virtual quadrotor UAV (quad-copter Simulation asset, Unity Assets) capable of flying in three dimensions was used to navigate the terrain. The operator could use arrow keys to move in front-back and left-right directions, and 'r' and 'f' keys to move up and down respectively. The UAV used realistic dynamics to simulate the motion while producing sound and rotor speeds in proportion to the thrust applied in each direction.

A camera fixed to the UAV provided the teleoperator a first person view (FPV) of the environment with UAV telemetry data superimposed on the display (Fig. 2). The camera could be pivoted to pitch up and down with 'w' and 's' keys providing a front or downward facing view of the virtual environment. The FPV display (Fig. 2) also showed an overhead GPS map of the environment in an inset on top right.

The inset showed the PLS and the three concentric rings denote high probability regions also called probability of areas (POAs) where the missing person may be found based on their walking speed. The radii of the POAs were set as the distance traveled by a simple random walker that changes their direction of motion randomly with speeds $s = \{1, 2, 3\}$ m/s in 36 h. Accordingly, the average distance was calculated as the square root of the variance of such a process over time ΔT $s\sqrt{\Delta T} = s\sqrt{(3600 \times 36)}$ [67, 68], where ΔT is the time for which the random walker is simulated. This translated to POAs with radii 0.36 km, 0.72 km, and 1.08 km centered at the PLS.

The missing person location was modeled after a random walker walking at a speed of 2 m/s and who stayed near the trails. Accordingly, the missing person was placed at a random location at about 0.72 km away from the PLS and within a radius of 100 m from the intersection of a trail. PLS locations were selected to be on trails that extended well beyond the largest POA radius in at least two directions. Although more sophisticated models of lost person movement that include age and mental fitness are available, we selected a simple model because the purpose of this study is to highlight the effects of prior knowledge in terms of where to search given the walking speed of a lost person only. The missing person asset was modeled as a male individual who wore a t-shirt and shorts. To further improve visibility and detectability, the missing person discharged an emergency flare, to send out red smoke signal that dissipated with distance.

The position of the teleoperated UAV was shown on the inset as a green circle with a yellow line pointing in the direction of heading. To give an appearance that other robots may also be searching the environment, fifty other robots were also placed randomly within in the environment and could be seen as red circles on the inset.

Experimental setup

The experimental setup consisted of a desktop computer with a wide-screen monitor (2560 × 1080 pixels, 60 Hz refresh rate, LG) to present the virtual environment, a fourteen-channel EEG device (Emotiv Epoc X, Emotiv Inc.) sampling at 256 Hz to record brain activity, and an eye tracking device recording gaze position at 150 Hz (Pupil Core, Pupil Labs inc.). Lab Streaming Layer software was used to record time-synchronized EEG, gaze, and keypress data. A chin rest was placed on the table at a fixed distance from the monitor to reduce head movement.

Experimental conditions

The experimental conditions consisted of varying the type of prior knowledge available and the spatial distribution of the robot swarm within the environment. Specifically, prior knowledge could be about the missing person in the form of their approximate distance from the PLS (two values, corresponding to whether such information was shown or not), terrain in the form of walking trails (two values, corresponding to whether they were shown or not), and spatial distribution of the robot swarm (two values, spread out uniformly or clustered on a random location within the outer

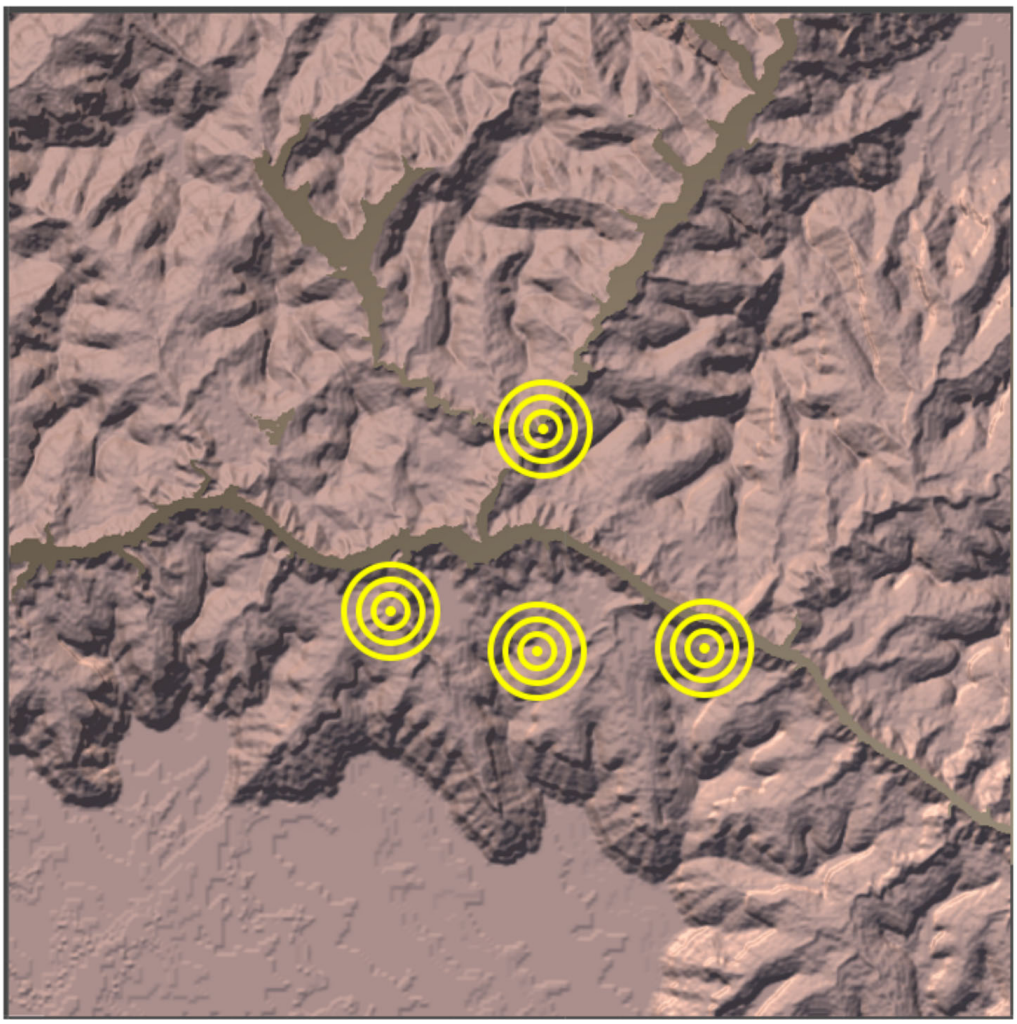


Fig. 1 An overhead view of the 10 km × 10 km virtual environment designed after the Grand Canyon National park using height map data. The four sets of concentric circles are centered at the four positions where the missing person could be last seen (PLS). Locations were selected on the basis of having equal number of trails and terrain similarity. The size of the circles correspond to different distances from the PLS based on how fast the missing person may have walked since they were last sighted

two POAs; the random location had no relation to the actual location of the missing person).

Correspondingly, we tested eight conditions, where each condition is encoded as a three-letter acronym depicting the presence of prior knowledge of missing person (Y/N), terrain (Y/N), and the distribution of swarm (U/C). For example, the condition NNU corresponds to where no prior knowledge was given about missing person or the terrain, and the UAV swarm were distributed uniformly through the environment; similarly the condition YYC corresponded to the prior knowledge provided for both the missing person and the terrain, and the UAV swarm clustered within the two outer POAs.

Experimental procedure

We recruited 20 participants (16 males and 4 females, aged 24 ± 4 years) for our study. (A power analysis for repeated measures within factors. with an effect size of 0.3, power of 0.95, and a significance criterion of $\alpha = 0.01$ and power = 0.95 revealed a minimum sample size of 18.) The participants were recruited through flyers and email announcements. Candidates were excluded if they were under 18 years of age, if they wore prescription glasses to prevent potential interference with eye-tracker, or if they had thick scalp hair which could prevent EEG headset from obtaining a reliable connection with the scalp. The experiments were approved by the institutional review board under protocol ID HS23-0233.

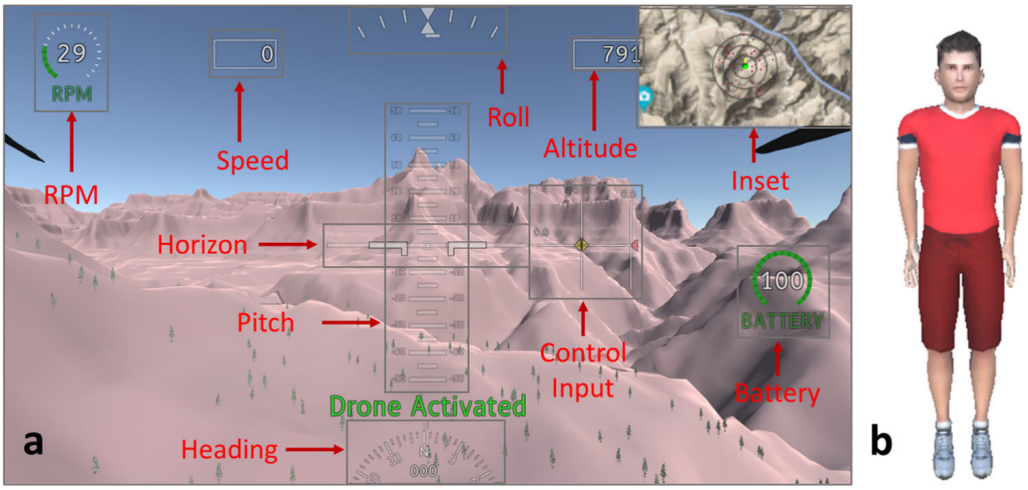


Fig. 2 FPV display (**a**) as seen by the teleoperator. The inset in the top right corner, shows the three POAs in black circles, the position of the teleoperated UAV (filled green circle), and other UAVs (red circles). The missing person (**b**) was placed within random locations within the virtual environment. See video of a sample search at https://youtu.be/3E6V3mOa7AM

Prior to the experiment, each participant was assigned a random six-digit number and asked to review and sign a consent form. The consent form listed the purpose of the study, potential risks, and the type of data that will be collected (EEG, eye tracking, and survey responses). Participants were then asked to put on the EEG headset using a reference image. The electrodes were then adjusted until a reliable connection was obtained. This was followed by instructions on how to teleoperate the UAV, details and purpose of the search mission, meaning of the POA circles, the different information shown on the FPV display, how to indicate having found the missing person, and how to recover from a crash. This was followed by a multiple choice questionnaire given to the participants to test their understanding. Participants were next asked to put on the eye tracker which was subsequently calibrated.

The experiment began with a familiarization trial followed by eight test trials. Participants were given a printed briefing prior to every trial including the familiarization trial. The briefing started with a paragraph describing the mission as "A person has gone missing while hiking the trails of the Grand Canyon..." and denoted that they were last seen 36 h ago at a location marked by a circular dot on a section of the map (Fig. 3). The briefing further added that the circles centered at the PLS have radii 0.36 km, 0.72 km, and 1.08 km. Information like UAV battery time and the possibility of seeing emergency flares near the missing person was also provided within the briefing. During the familiarization trial the experimenter helped the participant find the missing person by indicating their precise location on the map in the inset. Post familiarization, the eight conditions were tested in a random order. The search took place in one of the four locations as shown in Fig. 1. We note that the four locations were not counterbalanced between conditions as they were adjudged to be similar in terms of the terrain features. All participants were tested equally on all four locations so that each participant saw each location twice, once each for clustered and unclustered swarm for a given condition.

During trials, additional information regarding the missing person and the terrain was provided depending on the condition being tested. Specifically, for missing person information the briefing contained italicized text that said "Based on additional information about the person, physical and mental health, we think that the person may have walked between 0.36 km and 0.72 km (between smallest and the next-sized circle) during the last 36 h". For terrain information, the briefing contained text "Nearby trails are shown in black" on the printed map within the briefing. Figure 3 shows the two types of maps provided to participants.

Trials ended when participants either found the missing person or at approximately 10 min when the battery got depleted. Upon finding the missing person, the participant was asked to enter the color of the missing person's t-shirt and pants (one of four possible combinations) in a prompt. This was to ensure that the UAV was brought close enough for identification. At the end of every trial, the participant was asked to fill out a NASA-TLX survey to assess their mental workload on a 21-point scale. The participants were also offered to take a break after the fourth trial.

At the end of the experiment, participants were asked to enter their preferred gender and their age, followed by a presence questionnaire which consisted of questions on a 7-point scale. The questions surveyed the participants on their past experience with virtual reality, realism of the environment, responsiveness, where they tended to search for the missing

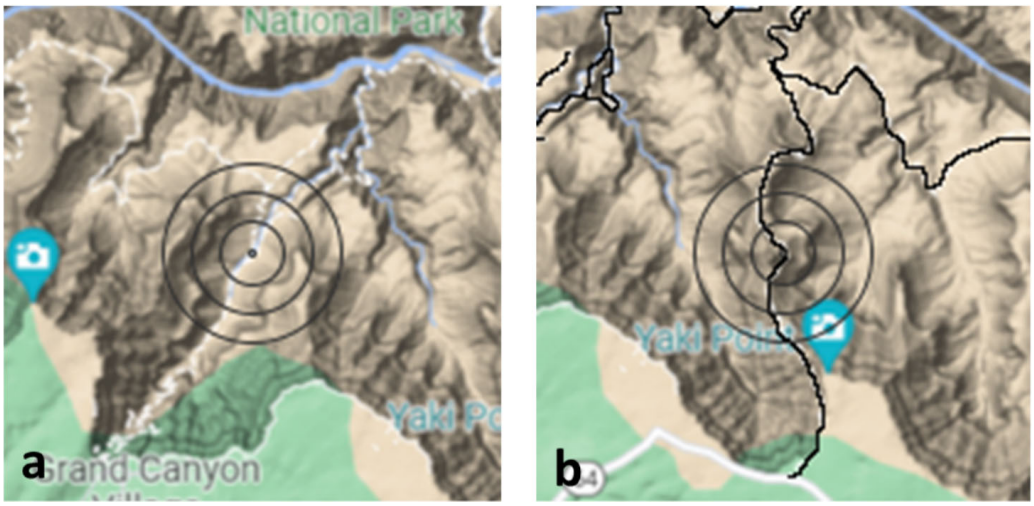


Fig. 3 Overhead views shared with the participants as part of their briefings. The views obtained from Google maps were overlaid with the POAs (grey concentric circles), for conditions with no terrain knowledge (**a**), and with terrain knowledge (**b**)

person, and whether the presence of other robots affected their search.

Data analysis

Calculating movement features Trajectory data from the experiments was processed to calculate four types of movement features: speed, turn rate, fraction of time staying still (freezing), and fraction of time spent turning while staying in place (turning while still). The trajectory data was smoothed with a moving window average of 0.5 s and used to calculate speed and turn rate of the UAV, freezing time and turning while still time. These features were selected as they can be reliably measured from the pose observation in the field for movement based inference of knowledge and cognitive state; freezing and turning while still behaviors are expected to additionally serve as indicators of situational awareness with a searcher possibly stopping to assess where they are in the environment and the mission [69].

Given the position $\mathbf{r}(k) \in \mathbb{R}^3$ of the UAV at kth time step, the instantaneous velocity of the UAV was calculated as $\mathbf{v}(k) = \frac{\mathbf{r}(k)-\mathbf{r}(k)}{t(k)-t(k-1)}$ where $t(k)$ is the corresponding timestamp. Speed $s(k) = \|\mathbf{v}(k)\|$, is the magnitude of velocity. Given the two-dimensional heading $\theta(k) \in \mathbb{R}$ of the UAV at kth time step, the instantaneous turn rate was calculated as $\omega(k) = \frac{\theta(k)-\theta(k-1)}{t(k)-t(k-1)}$. Freezing time was calculated as the fraction of the time over the duration of the trial when the teleoperated UAV was stopped completely, quantified as when $s(k) \leq 0.1$ m/s and $\omega(k) \leq 0.1$, and turning while still was calculated as the fraction of time when the UAV was in place and turning only, quantified as when $s(k) \leq 0.1$ m/s.

Calculating cognitive load and alpha-power from EEG data Cognitive load was calculated using both EEG and gaze data. With EEG data, cognitive load was calculated as the degree of reduction in α power denoted by frequencies within the 9.5–11.5 Hz range [53, 70]. In particular, EEG data was bandpass filtered to keep frequencies between 0.1 and 20 Hz followed by rejection of trials where any of the frontal electrodes posted an absolute amplitude above 1000 μV for greater than 5% of the length of the trial. This led to approximately 6% of the trials getting rejected. Post-filtering, for a given electrode k, the difference between the α power of baseline and trial was calculated as

$$\Delta I^k(\alpha) = I_b^k(\alpha) - I_t^k(\alpha), \tag{1}$$

where the baseline power $I_b^k(\alpha)$ is measured over data from 5 s prior to the start of the trial, and the trial power $I_t^k(\alpha)$ was measured over the length of the trial. Cognitive load was measured as the weighted average of this difference over all the frontal electrodes as

$$L(t) = \sum_{k=1}^{14} w_k \Delta I^k(\alpha), \tag{2}$$

where the weights w_k were based on an experimental study on object identification and classification [53]. Values of cognitive load that occurred beyond three times the standard deviation from the mean value across all participants were considered outliers and removed from the dataset.

Furthermore, α power has been shown to indicate level one situational awareness—paying attention to relevant elements of the scene. We accordingly calculated α power over an observation window o as $I_o(\alpha) = \sum I_o^k(\alpha)$.

Calculating fixations, dwell time, saccade frequency, and pupil diameter from eye tracking data The gaze location output from the eye tracker was measured in terms of normalized coordinates between 0 and 1, where the extreme

values corresponded to the lower left corner and upper right corner of the field of view bounded using QR codes. These values were mapped to computer screen pixel coordinates with a custom application designed in windows presentation foundation software. This application which ran prior to the experiment, asked the participants to fixate at the center of a cross-hair at nine different locations on the display. A second order polynomial was then fit to the normalized coordinates (x_i, y_i) at location i to output the pixel coordinates (X_i, Y_i) as

$$\begin{aligned} X_i &= a_0 + a_1 x_i + a_2 x_i^2 + a_3 x_i y_i \\ Y_i &= b_0 + b_1 x_i + b_2 x_i^2 + b_3 x_i y_i, \end{aligned} \tag{3}$$

where, $a_{0,1,2}$ and $b_{0,1,2}$ are constants found by solving the resulting system of equations.

A dispersion-threshold identification (I-DT) algorithm was used to quantify gaze fixation from the calibrated gaze data [71]. The I-DT algorithm identifies a group of consecutive gaze points as fixation based on an angular dispersion (how far the gaze moves during a set of consecutive points in time) and a minimum duration (how long the gaze stays within a particular region). In this study, we selected a dispersion threshold of 1° and a minimum duration threshold of 100 ms.

In our case, the dispersion threshold in degrees was converted to 8.5 mm of arc on the computer monitor using the distance of chin rest from the monitor, which was then used to determine fixation events. Consecutive gaze locations were marked as fixation events if the traverse distance was less than the dispersion threshold, and cumulative time exceeded the minimum duration threshold. The output of the algorithm was individual fixation event duration and its mean location. This was used to then calculate dwell time defined as the fraction of duration of the fixation events inside an area of interest (AOI) over the duration of the full experiment. In this study, the AOI comprised the full screen with the three corners (top left, bottom left and right) having the size of the inset map excluded (See supplementary material for a figure showing the AOI). The event between two fixation events is a saccadic event, the rate of such saccadic events is was calculated as saccadic frequency.

Pupil diameter measured in mm was thresholded to reject values above 9 mm. Further, average baseline measurement of pupil diameter calculated over the initial 5-second period before the trial was subtracted from the experimental values.

Statistical analysis

Statistical analysis was performed to assess movement correlates of prior knowledge, cognitive load and situational awareness as follows. For prior knowledge, generalized linear mixed models (GLMM) were fit to determine the main effects of swarm distribution, missing person knowledge, and terrain knowledge on speed, turn rate, freezing time, and turning while still. A significant effect was considered if the p value was less than 0.05. If found significant, the slope was then noted to comment on the size of the effect. Next, to motivate real-time inference of cognitive load and situational awareness from movement data, we calculate correlations of each with movement features over limited-time observation window. Specifically, for cognitive load, speed, turn rate, freezing time, and turning while still over the length of an observation window was correlated with the weighted difference of α power as calculated from EEG data, pupil dilation and saccade frequency for different observation windows lasting 5, 10, 15, and 20 s. For situational awareness, along the same lines, speed, turn rate, freezing time, and turning while still over the length of an observation window was correlated with gaze dwell time and α power for different observation windows.

Results

Realism of the virtual environment

Participants' responses to questions related to the virtual environment revealed that they found the environment to be responsive and comfortable and that they were generally eager to explore the virtual environment (Table 1). In regards to operating the UAV, they found it was controllable, felt its movement to be natural, and did not experience motion sickness. Notably, participants felt that the presence of other robots did not affect their search.

Sample trajectories

Figure 4 shows sample data from a participant including their speed, turn rate, gaze, and cognitive load calculated over a one-second segments for four out of eight trials. In general, we found that participants took about 20 s to start flying faster and actively searching. Participant gaze appears clustered either around the top right section of the display, which corresponded to the inset, or near the extreme right which corresponded to the battery level indicator.

Workload based on NASA-TLX survey

Table 2 shows the main effects of prior knowledge and swarm distribution on the responses to NASA-TLX survey questions that were posed to participants after every session (mean and standard deviation of responses to all questions reported in Supplementary material). The only response that was found to be significantly affected by prior knowledge was performance, where participants perceived as having performed better when they possessed prior knowledge for missing person; although marginally significant, participants also felt less frustrated when they had prior knowledge about the missing person.

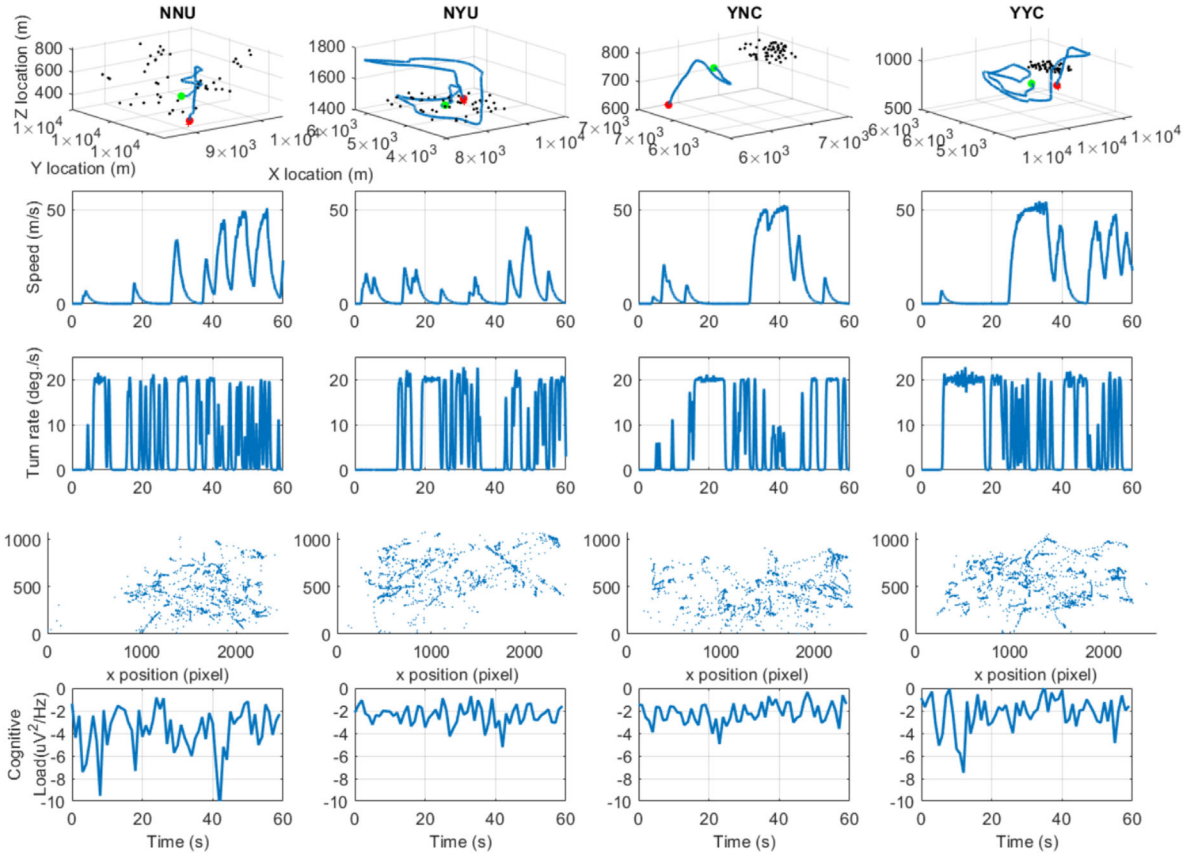


Fig. 4 Sample data of a single participant showing (in rows from top to bottom) the trajectory of the teleoperated UAV, speed, turn rate, gaze, and cognitive load (calculated on an observation window of 1 s), of the teleoperated UAV. Trajectory is shown for the full length of the experiment, whereas speed, turn rate, and gaze are shown for the first sixty seconds only. In the top row, green dot marks the start position and red dot marks the missing person location; black dots mark the location of autonomous UAVs held in position

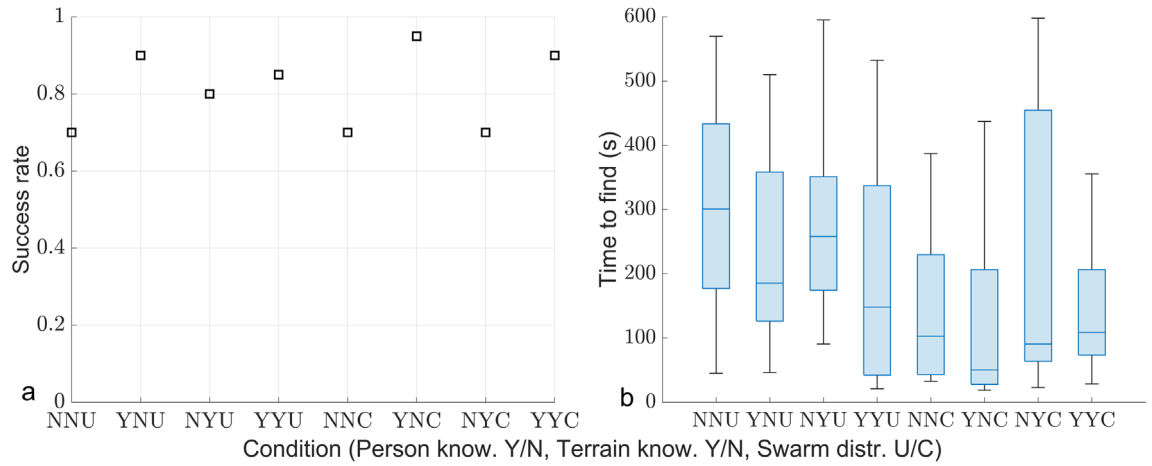


Fig. 5 Success rate (**a**) and time-to-find (**b**) the target among successful trials across the experimental condition

Table 1 Participant response to post-experiment questionnaire

	Response (mean ± std)
Questions about virtual reality (0–6)	
How would you describe your past experience with VR? (rare—frequent)	1.70 ± 1.42
How responsive did you find the virtual environment? (not responsive—very responsive)	3.40 ± 1.19
How comfortable did you feel? (very uncomfortable—very comfortable)	3.80 ± 1.54
How interested were you to explore the virtual environment? (not interested—very interested)	3.80 ± 1.47
Questions about operating the UAV (0–6)	
How natural did you find the movement of the quad-rotor (UAV)? (artificial—natural)	3.30 ± 1.63
How well could you fly through the environment? (with difficulty—with ease)	4.00 ± 1.17
The extent to which you felt as if you were moving when standing still? (not at all—very much)	1.95 ± 1.70
Did the presence of other UAVs affect your search? (did not—affected significantly)	1.10 ± 1.60

Table 2 GLMM estimates of responses to NASA-TLX questions as a function of the three independent variables

Predictor (min, max value)	Estimate	p	Predictor (min, max value)	Estimate	p
Mental (1, 21)			**Performance** (1, 21)		
Intercept	7.4	–	Intercept	10.987	–
Swarm distr (U, C)	− 0.672	0.441	Swarm distr (U, C)	0.8864	0.421
Prior knowledge, missing person (N, Y)	− 1.446	0.098	Prior knowledge, missing person (N, Y)	**4.036**	<0.001
Prior knowledge, terrain (N, Y))	− 0.153	0.86	Prior knowledge, terrain (N, Y))	0.1386	0.899
Physical (1, 21)			**Effort** (1, 21)		
Intercept	3.337	–	Intercept	8.787	–
Swarm distr (U, C)	0.1	0.881	Swarm distr (U, C)	− 1.198	0.186
Prior knowledge, missing person (N, Y)	− 0.4	0.551	Prior knowledge, missing person (N, Y)	− 1.448	0.11
Prior knowledge, terrain (N, Y))	− 0.475	0.478	Prior knowledge, terrain (N, Y))	− 0.176	0.845
Temporal (1, 21)			**Frustration** (1, 21)		
Intercept	6.3	–	Intercept	5.562	–
Swarm distr (U, C)	− 1.085	0.151	Swarm distr (U, C)	− 0.153	0.857
Prior knowledge, missing person (N, Y)	− 1.061	0.16	Prior knowledge, missing person (N, Y)	− **1.627**	0.056
Prior knowledge, terrain (N, Y))	− 0.014	0.985	Prior knowledge, terrain (N, Y))	0.277	0.743

Values in bold indicate significant factors

Table 3 GLMM estimates of mission performance as a function of various independent factors

Predictor (min, max value)	Estimate	p
Performance (0, 600 s)		
Intercept	285.468	< 0.001
Swarm distr (U, C)	− 90.931	**0.001**
Prior knowledge, missing person (N, Y)	− 66.66	< **0.016**
Prior knowledge, terrain (N, Y)	2.818	0.917

Values in bold indicate significant factors

Search performance

Figure 5 shows the search performance in terms of the time-to-find the target across different conditions. The time-to-find values exclude participants who were not successful in finding the target. Success rates are 90% or above when person knowledge is available and 70% or below when neither person nor terrain is available except NYC where we see a low success rate even when terrain knowledge is available. Median values of time-to-find suggest that participants perform better in successful trials when they possessed knowledge about the missing person compared to when they possessed only terrain knowledge. For example, the median performance of YNC is better than NYC, YNU is better than NYU. At the same time, when looking at the distribution of two-dimensional positions for four different combinations of prior knowledge (Fig. 6), we see that participants did follow trails during their search when they were available (NY and YY) ; participants tended to follow the river in the (NN) condition that happened to align with the trail.

With respect to time-to-find, we note a trend in terms of swarm distribution where a clustered swarm (that was located

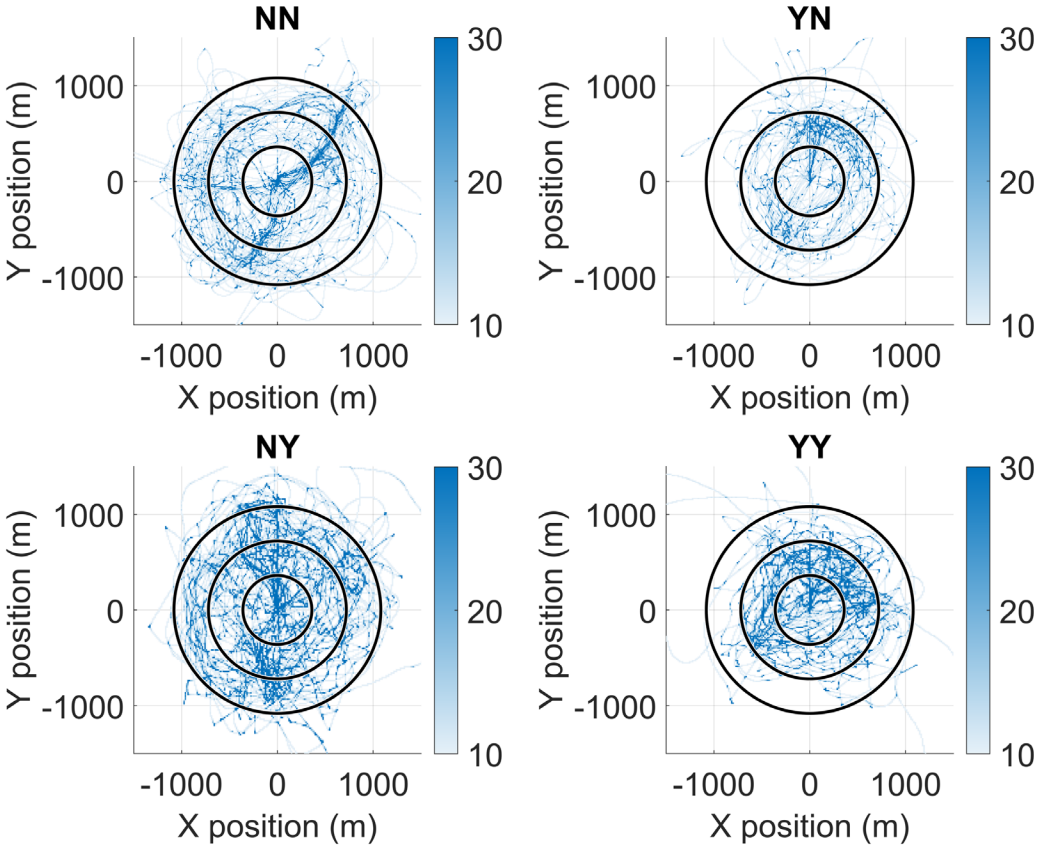


Fig. 6 Distribution of teleoperator trajectories in a 2D plane for the four different locations after combining the the data for clustered and unclustered swarms. The corresponding search locations with trails are shown in Supplementary document. Here, NN represents no person knowledge and no terrain knowledge, and NY represents no person knowledge by yes terrain knowledge

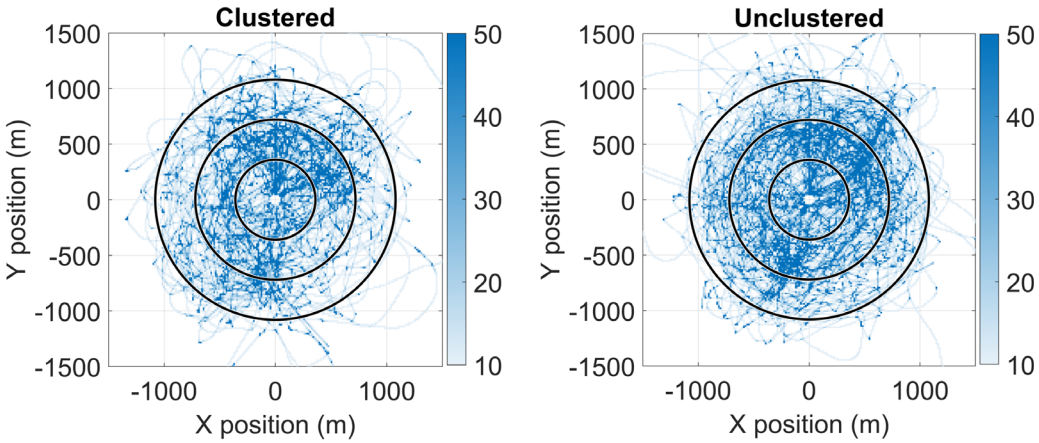


Fig. 7 Distribution of teleoperator trajectories in a 2D plane across all locations but divided in terms of whether the swarm of robots was clustered or unclustered

Table 4 GLMM estimates of UAV movement features as a function of various independent factors

Predictor (min, max value)	Estimate	p
Average speed (0, 50 m/s)		
Intercept	27.405	< 0.001
Swarm distr (U, C)	− 3.152	**0.0293**
Prior knowledge, missing person (N, Y)	− 1.488	0.303
Prior knowledge, terrain (N, Y))	1.461	0.3031
Average turn rate (0, 25 °/s)		
Intercept	6.165	< 0.001
Swarm distr (U, C)	− 0.555	0.174
Prior knowledge, missing person (N, Y)	− 0.864	**0.036**
Prior knowledge, terrain (N, Y)	− 0.262	0.519
Freezing time (0, 1)		
Intercept	0.032	< 0.001
Swarm distr (U, C)	0.013	0.091
Prior knowledge, missing person (N, Y)	0.012	0.132
Prior knowledge, terrain (N, Y)	− 0.007	0.343
Turning while still (0, 1)		
Intercept	0.067	< 0.001
Swarm distr (U, C)	0.015	0.174
Prior knowledge, missing person (N, Y)	0.002	0.823
Prior knowledge, terrain (N, Y)	− 0.019	0.099

Values in bold indicate significant factors

randomly in the environment) led to improved performance than a swarm that was distributed throughout the region.

Table 3 lists the results of GLMM on the effect of experimental conditions on time-to-find and confirms these trends. We find a significant effect of swarm distribution with a clustered swarm leading to lower times-to-find. We also find significantly lower time-to-find when missing person knowledge was available. To investigate the dependence of performance on swarm distribution further the teleoperator trajectories in the 2D plane are shown for the clustered and unclustered swarms (Fig. 7). Here, we see that participants tended to stay more within the first and second POAs when the swarm was clustered compared to when it was not.

Movement dependence on prior knowledge

Figure 8 compares the four different movement features across experimental conditions. All values were calculated over the duration of a session. Table 4 shows the main effects of the independent factors in the experiment on the same movement features. We find that average speed of the teleoperator decreases when a swarm is clustered, and average turn rate decreases when missing person knowledge is available.

Movement correlates of cognitive load and SA

Table 5 lists correlations between measures of cognitive load, situational awareness and the four movement features over different observation windows. Measures of cognitive load include EEG-based weighted average of difference between α power (2), pupil diameter, and saccade frequency. Measures of situational awareness include dwell time within AOI and α power. All these measures were calculated and correlated with movement features over 5, 10, 15 and 20 s windows.

Most cognitive load measures show no or very weak correlation with movement features except saccade frequency which shows a weak correlation with turn rate at a 5, 10, and 20 s windows but not at 15-second window. Visual inspection of plots of movement features with respect to cognitive load measures of EEG, pupil dilation and saccadic frequency did not reveal and nonlinear or threshold-based relationships. Among measures of situational awareness α power shows a weak correlation with turning while still behavior over all four second observation windows.

To summarize the experimental results, we find the following:

1. Participants found the virtual environment realistic and the movement of the quad-rotor responsive
2. Knowing how far the missing person may have traveled improved perceived performance and lowered frustration levels of participants as reported in a workload questionnaire.
3. Performance also depending on swarm distribution although participants indicated that presence of other robots did not affect their search.
4. Performance of search in terms of success in finding the person within 10 min is high when prior knowledge of missing person and terrain is available and low when none of them are available. Performance in terms of time-to-find the missing person is high when prior knowledge of missing person is available and swarm distribution is clustered.
5. In terms of teleoperated robot's movement cues that are affected by prior knowledge (H1) we found that only turn rate was significantly affected with the teleoperating UAV having lower turn rates when the participants had missing person knowledge. Terrain knowledge did not significantly affect any of the movement cues.
6. Whereas cognitive load measured using EEG showed very weak correlation to speed, freezing and turning while still, cognitive load measured using saccadic frequency it showed weak correlation to turn rate (H2); SA measured using α power showed weak correlation to turning while still (H3).

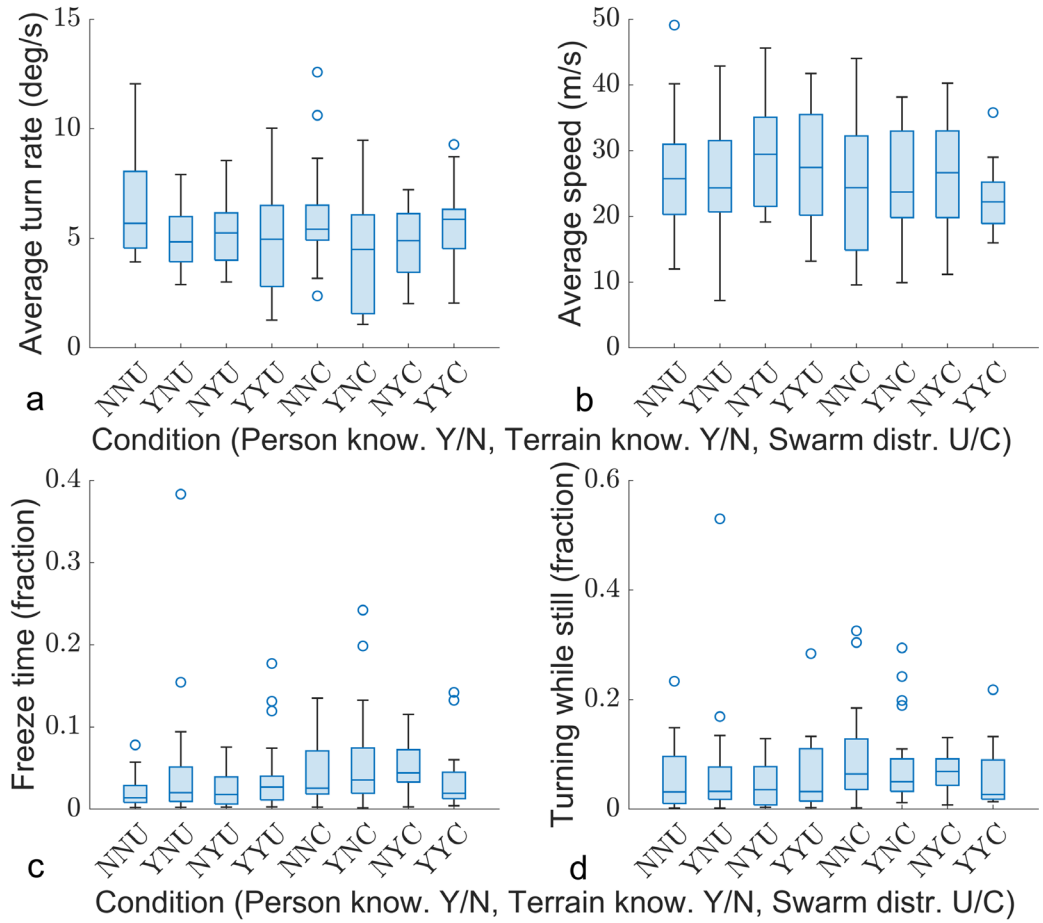


Fig. 8 Plots showing **a** average turn rate during trial, **b** average speed during trial, **c** freezing time, **d** and time turning while still across experimental conditions

Estimating human prior knowledge and state based on movement cues

Based on our experimental results, that show strong dependence of turn rate on prior knowledge and weak dependence of cognitive load and situational awareness on teleoperated UAV movement, we next build classifiers for estimating these values directly from movement features. In particular we use LSTM to build a prior knowledge classifier and linear regression to build a situational awareness estimator; we do not build a classifier for cognitive load because of the inconsistency in slope with respect to observation window size suggesting that such an estimator may not be robust. Notably, α power which was found to correlate with turning while still is an acceptable measure of cognitive load [70] and therefore indirectly serves to capture both situational awareness and cognitive load.

Estimating prior knowledge

We built an LSTM network to estimate the probability of having prior knowledge about the missing person and terrain based on the movement of the teleoperator. The model was trained on over 12 h of trajectory data collected during the experiment.

The LSTM was setup as a multi-label classifier, where the output of each type of prior knowledge (person or terrain) was between 0 and 1, with 0 signifying no prior knowledge and 1 being full prior knowledge. To design a meaningful estimator which can be used for near real time response in the field, as with situational awareness and cognitive load, we sought to classify prior knowledge based on limited observations. Accordingly, the trajectory data was partitioned into sections representing observation windows, and the speed and turn rate data over that section served as an input to the model. Hyperparameter sets included observation window size of 5, 10, 15, 20 seconds and mini-batch sizes of 256, 512. These were varied to maximize accuracy of the model defined as the number of correct predictions over true pre-

Table 5 Correlation coefficient (with p-value) of cognitive load and situational awareness with different movement features of turn rate, speed, freezing time and turning while still

Window size (s)	5	10	15	20
Turn rate (deg./s)				
α-difference	0.011 (0.350)	0.021 (0.169)	0.024 (0.189)	0.028 (0.202)
Pupil diameter	0.002 (0.858)	0.004 (0.812)	0.0063 (0.743)	0.0077 (0.731)
Saccade frequency	**− 0.138 (<0.001)**	**− 0.142 (<0.001)**	**− 0.068 (<0.001)**	**− 0.158 (<0.001)**
Dwell time	**0.045 (<0.001)**	**0.056 (<0.001)**	0.058 (0.002)	**0.073 (<0.001)**
α-power	0.011 (0.305)	0.028 (0.063)	0.017 (0.367)	0.012 (0.594)
Speed (m/s)				
α-difference	**− 0.038 (<0.001)**	**− 0.047 (0.002)**	**− 0.053 (0.004)**	**− 0.068 (0.002)**
Pupil diameter	− 0.0023 (0.833)	− 0.0014 (0.925)	0.003 (0.877)	0.0077 (0.8063)
Saccade frequency	**− 0.036 (<0.001)**	**− 0.067 (<0.001)**	− 0.014 (0.475)	**− 0.074 (0.002)**
Dwell time	0.0158 (0.145)	-8×10^{-4} (0.958)	− 0.022 (0.243)	− 0.039 (0.0793)
α-power	− 0.017 (0.107)	− 0.0391 (0.010)	− 0.033 (0.073)	− 0.041 (0.057)
Freezing time (fraction)				
α-difference	0.015 (0.148)	**0.045 (0.002)**	**0.058 (0.002)**	**0.083 (<0.001)**
Pupil diameter	− 0.0034 (0.755)	− 0.0034 (0.755)	0.0042 (0.829)	− 0.0037 (0.869)
Saccade frequency	0.033 (0.0025)	0.036 (0.022)	− 0.012 (0.538)	− 0.077 (0.001)
Dwell time	**− 0.055 (<0.001)**	**− 0.06 (<0.001)**	**− 0.062 (0.001)**	− 0.043 (0.049)
α-power	**0.068 (<0.001)**	**0.073 (<0.001)**	**0.083 (<0.001)**	**0.082 (<0.001)**
Turning while still (fraction)				
α-difference	0.0255 (0.0168)	**0.0601 (<0.001)**	**0.0807 (<0.001)**	**0.1197 (<0.001)**
Pupil diameter	0 (0.965)	0.0012 (0.941)	0.001 (0.957)	− 0.0016 (0.943)
Saccade frequency	0.006 (0.572)	**0.016 (0.022)**	**− 0.028 (0.133)**	**0.065 (0.006)**
Dwell time	**− 0.074 (<0.001)**	**− 0.0797 (<0.001)**	**− 0.074 (<0.001)**	− 0.0546 (0.013)
α-power	**0.108 (<0.001)**	**0.114 (<0.001)**	**0.127 (<0.001)**	**0.136 (<0.001)**

Cognitive load was measured using α-difference, pupil diameter, and saccade frequency, and SA was measured using dwell time and α-power
Values in bold indicate significant factors

dictions on a validation dataset. Here, for example, a correct prediction is counted if the probability for both types of prior knowledge (person and terrain) matched that of the experimental condition; a probability above 0.5 indicated a correct inference of the presence of prior knowledge.

The neural network architecture selected for the LSTM was based on [72] which has been shown to work well for time-series classification. The original model was adapted to enable two input features (speed and turn rate), and a multi-label classification was used instead of multi-class classification. To optimize the model, we used a variation of k-fold cross validation to find the optimal hyperparameters (see algorithm in Supplementary material). Figure 9, shows the neural network architecture used in this study. Figure 10 shows the maximum accuracy obtained as a function of observation time. Based on this, we selected an observation of 15 s and a mini-batch size of 512 that produced the maximum accuracy of 0.7 ± 0.07.

Estimating situational awareness

To estimate SA from a movement cue we select α power and fraction of time spent turning while still. In particular, we used linear regression to estimate α power from the fraction of time spent turning while still over an observation window of 15 s selected to match the observation window for estimating prior knowledge. Denoting the fraction of time spent turning while still by f_{turn}, we estimate α power for 15 s, $I_{15}(\alpha)$, as

$$I_{15}(\alpha) = 0.34698 + 11.228\, f_{turn}, \tag{4}$$

where 0.3468 is the intercept and 11.228 is the slope. The estimated α power is then normalized so that it lies between 0 and 1 as

$$\hat{I}_{15}(\alpha) = \frac{I_{15}(\alpha) - I_{15}(\alpha)_{min}}{I_{15}(\alpha)_{max} - I_{15}(\alpha)_{min}}, \tag{5}$$

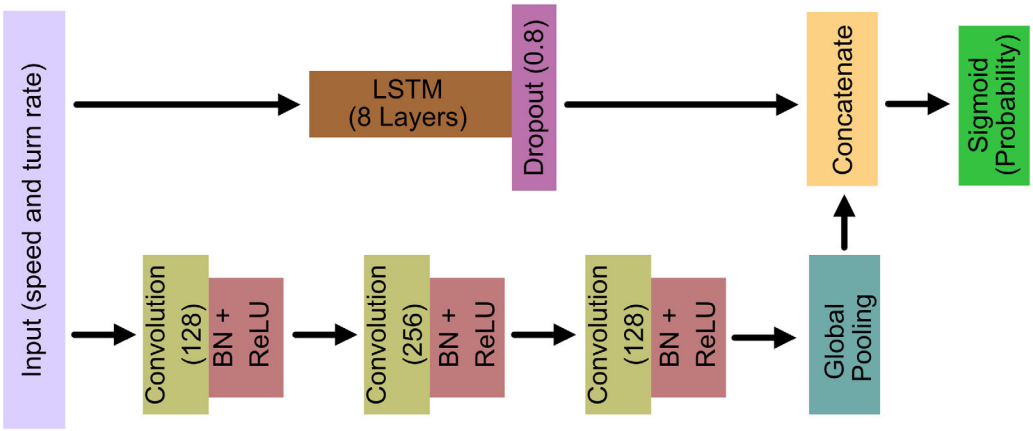


Fig. 9 The LSTM neural network architecture used for classifying prior knowledge. The architecture consists of blocks made of 2D convolution layer followed by batch normalization (BN) and a rectified Linear Activation function (ReLU). The input to the LSTM network are turn rate and speed measurements of the human teleoperated UAV over a limited-time observation window and the output is probability of having person or terrain knowledge (available as part of the experimental conditions)

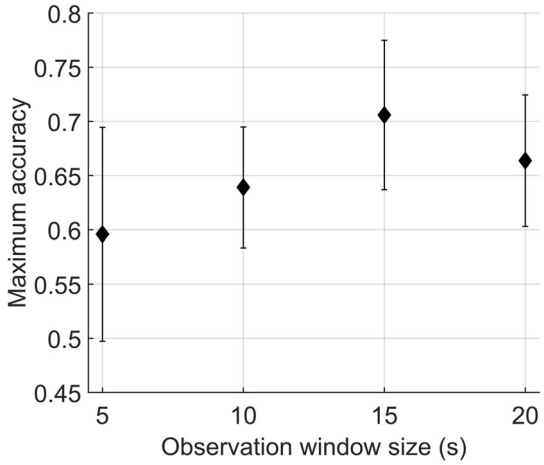


Fig. 10 Maximum accuracy of the LSTM classification of prior knowledge based on movement cues as a function of observation window size. Mean and standard deviation represents values over a k-fold validation procedure with 5 runs

where $I_{15}(\alpha)_{max}$ and $I_{15}(\alpha)_{min}$ are calculated over all the trials.

Adaptive search based on prior knowledge and SA

In this section we formulate adaptive search strategies for an autonomous robot swarm that assists a human teleoperator in search. The robot swarm is informed by the estimated prior knowledge and SA of the human teleoperator within a human–robot team. These strategies are then compared with random search and spiral search in terms of success rate, time-to-find the missing person, and control effort.

Adaptive search strategies

We modeled the movement of the autonomous robot swarm according to a modified version of the kinematic self-propelled particle model called the zonal model [65]. Specifically, the position $\mathbf{r}_i(t) \in \mathbb{R}^3$ and velocity $\mathbf{v}_i(t) \in \mathbb{R}^3$ of i-th autonomous robot at time t is updated to (a) avoid collisions, (b) align with direction of motion of neighbors, and (c) remain within a certain range of the swarm, all while preferentially following the human operator. Accordingly, the direction of motion of a robot i is updated as

$$\mathbf{d}_i(t+\delta t) = \begin{cases} -\mathbf{d}_{i,zor}(t) \text{ if } zor_i \neq \emptyset \\ [\mathbf{d}_{i,zoo}(t) + \mathbf{d}_{i,zoa}(t)](1-k_h(t)) \\ \quad +k_h(t)\mathbf{d}_{i,h}(t) \text{ otherwise,} \end{cases} \tag{6}$$

where (dropping time dependence for clarity),

$$\mathbf{d}_{i,zor} = \sum_{j \in zor} -\frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \ \mathbf{d}_{i,zoo} = \sum_{j \in zoo} \frac{\mathbf{v}_j}{|\mathbf{v}_j|},$$
$$\mathbf{d}_{i,zoa} = \sum_{j \in zoa} \frac{\mathbf{r}_i - \mathbf{r}_j}{|\mathbf{r}_i - \mathbf{r}_j|}, \text{ and } \mathbf{d}_{i,h} = \frac{\mathbf{r}_h - \mathbf{r}_i}{|\mathbf{r}_h - \mathbf{r}_i|}$$

and $zoo(t)$, $zor(t)$, and $zoa(t)$ denote the sets of neighbors that lie within neighborhood zones marking a distance zoo, zor, and zoa from the focal robot i (slight abuse of notation so that the same symbol is used for the set and distance); the size of the zones $zoo = zor + 8$ m, and $zoa = zor + 22$ m, are selected to maintain a dynamic parallel behavior which entails that the swarm stays close together in a coordinated direction [65]. Indices i and j represent the focal robot and the neighboring robot respectively ($i \neq j$), h is the subscript for the human teleoperator robot, and $\delta t = 1/24$ s denotes

the simulation time step. The time-varying parameter $k_h(t)$ determines how strongly a robot follows the human teleoperator. The velocity of a robot $\mathbf{v}_i(t) = s_i(t)\frac{\mathbf{d}_i(t)}{\|\mathbf{d}_i(t)\|}$, where $s_i(t) \in \mathbb{R}$ denotes speed. The speed of the robots is matched to the speed of the human teleoperator in adaptive search and kept constant for spiral and random search strategies.

In a non-adaptive strategy the value of zor is set to match that of the field of view of the robot r_{fov} to (a) ensure minimal overlap of field of view between robots towards and efficient search and (b) prevent clustering.

To formulate adaptive search strategies, we propose the search actions by the autonomous swarm as listed in Table 6. Specifically, we consider an action for the four possible scenarios corresponding to the four experimental conditions in which the human prior knowledge can exist. For example, we justify an increased coverage of the terrain by the swarm combined with strong following of the human teleoperator when they have high missing person knowledge but no terrain knowledge.

To implement these strategies, we update the zor and k_h in (6) as a response to prior knowledge inference. To determine if utilizing person knowledge is enough or if terrain knowledge should also be used, we evaluate two types of strategies, one that uses both terrain and person knowledge inference, and another that uses person knowledge inference only. Furthermore, to evaluate the effect of adapting the search strategy in response to situational awareness, we regulate the parameter k_h by weighting it with normalized α power (from equation 5). This implies that following behavior in the swarm should increase with increase in situational awareness. Accordingly, we evaluate four adaptive search strategies involving human teleoperator and an autonomous swarm and two non-adaptive search strategies that involve the robotic swarm only:

a) **Search based on person and terrain knowledge (AsPkTk)** The value of zor is updated based on the terrain knowledge so that high terrain knowledge of the teleoperator leads to a smaller zor, which in turn should result in a tighter coverage of the region around the human teleoperator. Person knowledge directly affects the tendency of the swarm to follow the teleoperated UAV. Accordingly, we update zor and k_h as

$$\begin{aligned} zor(t) &= r_{fov}(\beta + e^{-\lambda T_k(t)}), \\ k_h(t) &= P_K(t), \end{aligned} \tag{7}$$

where $\beta = 0.9$ determines the minimum value of zor to be 90% of r_{fov} when maximum terrain knowledge is available, and $\lambda = 10.5$ determines the rate at which the zor decreases with increase in terrain knowledge.

b) **Search based on person knowledge only (AsPk)** Person knowledge directly affects the tendency of the swarm to follow the teleoperated UAV as

$$k_h(t) = P_K(t), \tag{8}$$

c) **Search based on person and terrain knowledge, and SA (AsPkTkSA)** The zor of the swarm is determined by terrain knowledge as with **AsPkTk** and the gain k_h is regulated by SA so that

$$\begin{aligned} zor(t) &= r_{fov}(\beta + e^{-\lambda T_k(t)}) \\ k_h(t) &= \hat{I}(\alpha) P_K(t), \end{aligned} \tag{9}$$

where as before $\beta = 0.9$ and $\lambda = 10.5$.

d) **Search based on person knowledge only with SA (AsPkSA)** person knowledge affects the gain k_h only which is regulated by SA as

$$\mathbf{k}_h(t) = \hat{I}(\alpha) P_K(t), \tag{10}$$

e) **Random search (Rs)** In this strategy, no human teleoperator is present and the autonomous robot swarm searches the region according to the zonal model [65] so that $k_h(t) = 0$. Additionally, a uniform random noise of $[-0.1, 0.1]$ rad/s is added to instantaneous turn rate of each robot. The speed of a robot is set to 26 m/s which approximates the average speed of the teleoperated UAV across all experiments.

f) **Spiral search (Ss)** In this strategy [73], the robot swarm follows a spiral trajectory achieved by using a modified version of (6) where the first agent of the swarm follows a predetermined path and the rest of the agents follow the first agent with $k_h(t) = 1$. The predetermined path is a spiral trajectory produced by placing successive waypoints along a spiral path designed to produce no more than 5 m overlap of coverage between successive arms. As with random search, the speed of a robot is set to 26 m/s.

Figure 11 shows the estimated probabilities of prior knowledge of person and terrain and normalized situational awareness inferred from movement cues calculated over a moving window of 15 s. The values correspond to the condition initially so that for example, the Person and terrain knowledge is low for NNU for the first 40 s before terrain knowledge rises. Terrain knowledge especially, also jumps between 0 and 1 several times mainly at the start of the trials.

Simulation setup

The search strategies were tested in the same virtual environment that was designed for experiment. The simulation environment replays back the trajectory of a participant from

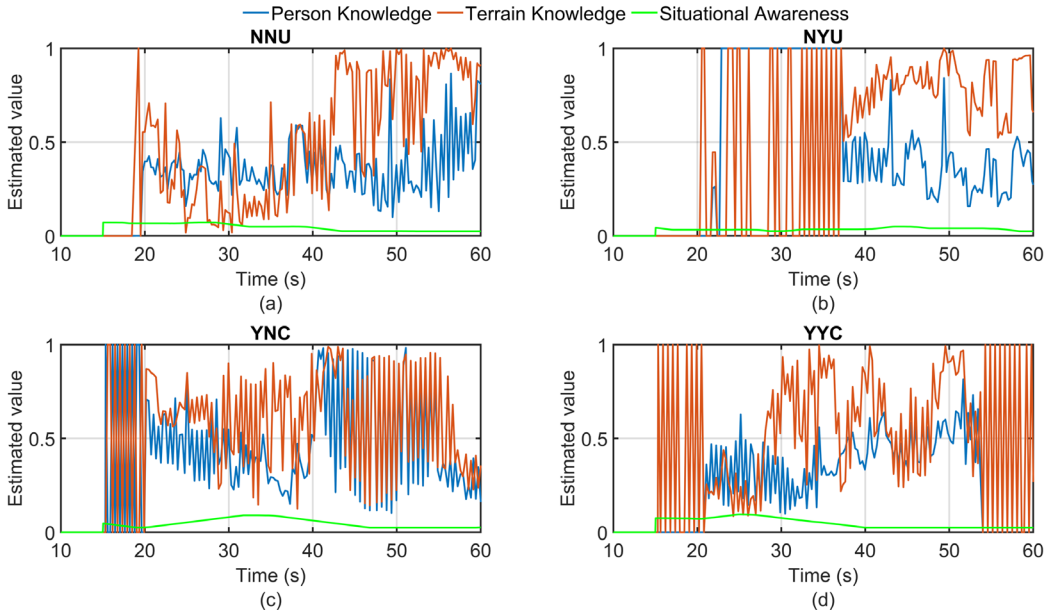


Fig. 11 Sample estimates of prior knowledge and SA of a participant with a 15-second observation window. Correspondingly, the value for the first 15 s is assumed zero

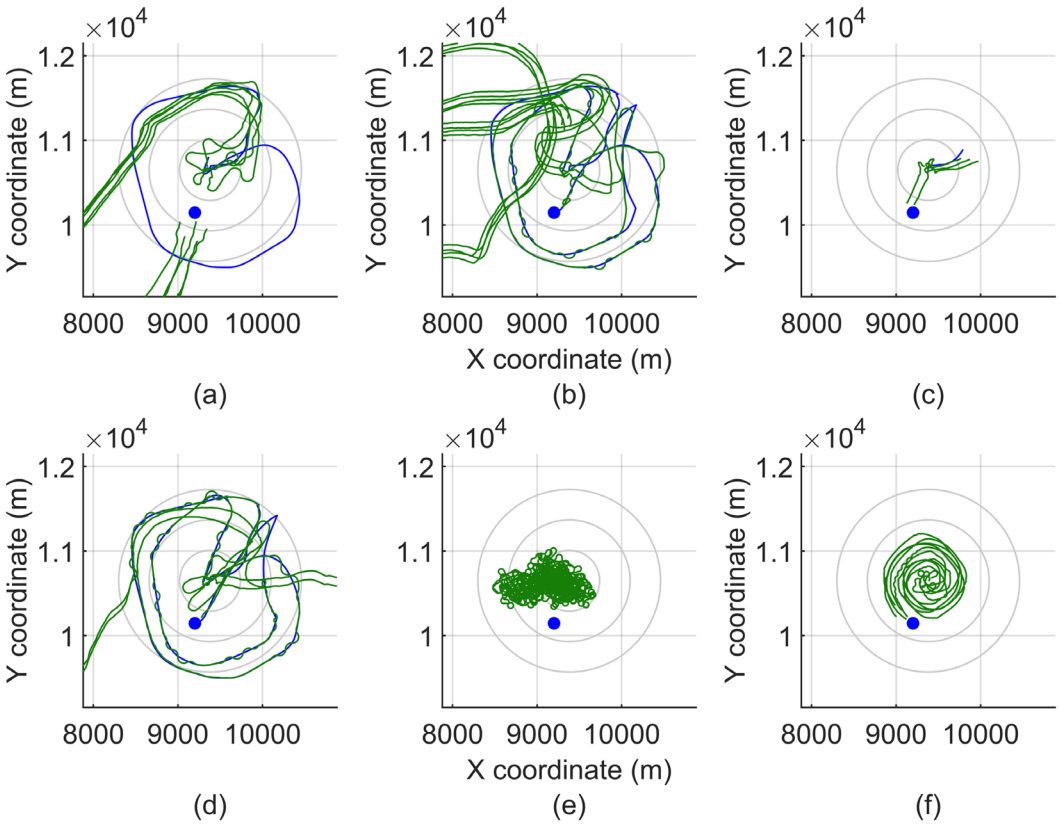


Fig. 12 Sample trajectories of teleoperated UAV (blue) and autonomous UAVs (dark green) in the order listed in " Adaptive search strategies", for the same condition. The grey circles represent the three POAs and the blue dot represents the location of the missing person

Table 6 Adaptive search strategies for prior knowledge scenarios corresponding to the experimental conditions

P_K	T_K	Action	Justification
1	0	Large separation of swarm, strong following	Increased coverage over the same terrain
0	1	Low swarm separation and strong following	Strong overlap in the same terrain because terrain knowledge
1	1	Large separation, very strong following	Increased coverage in the same terrain
0	0	Small separation, weak following	Not trusting the human operator, swarm searches randomly

the experiments as an autonomous swarm accompanies the search.

Simulations were run with a time step of 1/24 s. All autonomous robots flew at approximately 170 m above the lowest elevation of the map, with a field of view of about 30°. This corresponded to a $r_{fov} = 100$ m. The probability of detection of the target if it was located within the field of view of an autonomous robot was set to 0.6.

To encode the possibility of false positives, we visually analyzed the participant trajectories to count the number of instances that they descended their teleoperated UAV close to the terrain to investigate a possible target location. Excluding their final successful descents, we counted an average of two instances per participant. Accordingly, for each simulation, we randomly assigned two locations within a radius of 0.8 km from the PLS as false positives. If a robot came close to the false positive location then it would slow down by regulating it's speed according to a logistic function as

$$s_i(t) = \frac{s_0}{1 + e^{-a_s(d^{i,k}(t) - b_s)}}, \tag{11}$$

where s_0 is the initial speed of the robot, $d^{i,k}(t)$ is the distance of i-th agent from k-th false positive location, $k = 1, 2$ representing one of two locations of false positives, $a_s = 0.08$ which controls the rate of slowing down near a false positive and $b_s = 48.6$ which controls the distance at which slowing down occurs. In this sense, equation (11) represents a temporary slowing down of an autonomous robot near a false positive location as it comes within r_{fov} and resuming full speed as the distance to the false positive location increases. The slowing down near a false positive location is independent for all robots except for in the the spiral search strategy (Ss) where all robots slow down if any one of them come close to the false positive location. All search strategies were tested with 5, 10 and 20 robots.

Each simulation started with the autonomous robots placed randomly near the PLS with random directions of motion. Figure 12 shows sample trajectories of the autonomous robots for each of the search strategies. For adaptive search, the human teleoperator trajectory is also shown.

We test the strategies against four experimental conditions corresponding to the four types of prior knowledge (NN, YN, NY, and YY, for person and terrain). These correspond to 40 trials where a human teleoperator trajectory is available after combining the trials for clustered and unclustered swarm condition in the experiments. Random and spiral search are also run for 40 instances for each of the conditions.

A search by a human-swarm team is considered successful if any of the team members (human teleoperated or autonomous) finds the missing person. The missing person is considered found if their location falls within the r_{fov} of an autonomous robot with a detection rate of 60% and before 600 s have passed. To compensate for the time it took for a participant to descend and enter in the identifying information in the experiments, the missing person was considered found by the human teleoperator 15 s prior to the finish of the experimental trial.

Simulation results

Figure 13 shows the results of simulating different search strategies for each of the four types of prior knowledge that a human teleoperator can possess in our experiments. We find that nearly all adaptive search strategies by the human–robot team achieve higher success rate than random search or single human search. Among the different search strategies, all of them have a success rate of more than 80% with even higher success rates of above 90% when missing person knowledge was available. The spiral search strategy almost always performed at a 100% success rate.

When comparing average time-to-find (Fig. 14), contrastingly, the adaptive search strategies tend to perform better than spiral search with consistently lower times-to-find across all the strategies. This difference between times-to-find is amplified to more than 150 s (approximately 60% lower) with larger swarms of 15 robots in some cases when person knowledge is not available to the teleoperator (e.g. NY). The effect of having an autonomous swarm assist the human teleoperator is visible both in terms of achieving higher success rate and lower times-to-find.

In terms of which strategy among the four is helpful, when only looking at the five-robot swarm, we see that strategy AsPkSA had the lowest success rates among all strategies while AsPkTkSA had the highest success across all conditions closely followed by AsPk. When comparing times-to-find, all strategies performed similar except the

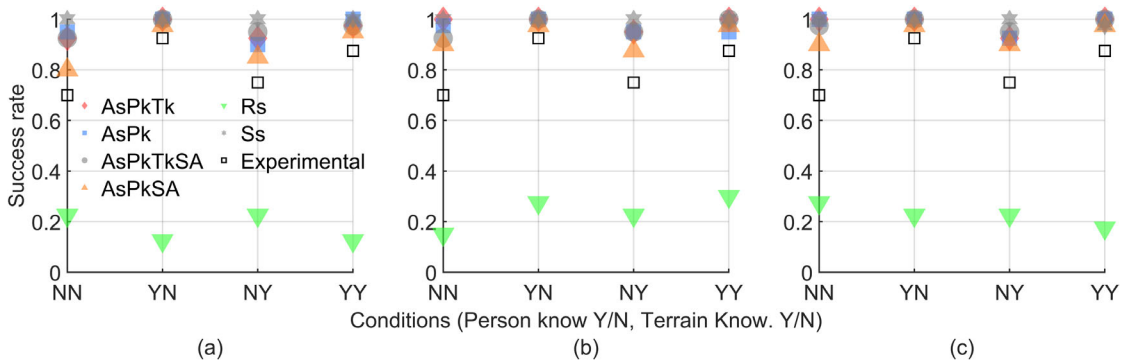


Fig. 13 Success rate of finding the missing target by a human–robot team (for strategies **a**–**d**) and swarm (strategies **e** and **f**) with 5 (**a**), 10 (**b**) and, 15 (**c**) autonomous robots. Success rate of a single human from experiments (Experimental) is shown for reference

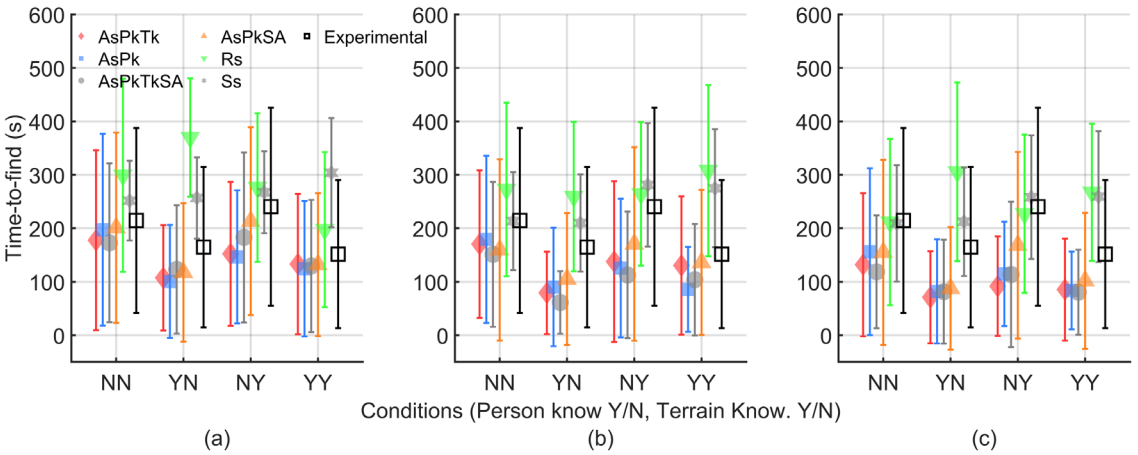


Fig. 14 Average time-to-find the missing target for successful simulations by human-swarm team (strategies **a**–**d**) and swarm only (strategies **e** and **f**) with 5 (**a**), 10 (**b**), and 15 (**c**) autonomous robots. Average time-to-find by a single human from experiments (Experimental) is shown for reference

AsPkSA which had slightly higher time-to-find when person knowledge was not available but terrain knowledge was (NY).

With respect to control effort, we calculated average magnitude of acceleration for a five-member swarm as an indirect measure of control effort. We find that the adaptive strategies use significantly more effort than random search or spiral search (see Supplemetary material).

In terms of the number of autonomous robots, while we don't see any improvement in random search, we do find that with 15 robots, nearly all strategies perform at close to or above 95% success rate.

Discussion

Executing a SAR mission with a teleoperated UAV involves taking multiple time-critical decisions. In such a scenario, prior knowledge about the mission would naturally play an important role. Additionally, performance and the effect of prior knowledge are likely regulated by the cognitive load and situational awareness of the teleoperator. Quantifying these effects on movement has implications in shared autonomy where autonomous robots can "read" the operator movement in the field and assist accordingly. Our experimental results point to significant albeit sometimes weak effects of prior knowledge, cognitive load and SA on movement. We build on these results to create movement-based classifiers of prior knowledge and situational awareness and design new adaptive search strategies for human–robot team based search. Simulation results with human-in-the-loop experimental trajectories show that adaptive search strategies have high success rates compared to random search and single human and lower times-to-find compared to spiral search.

Perception of the experimental platform and NASA-TLX reported workload

In terms of workload as measured using NASA-TLX, the tasks were not perceived to be physically demanding, and participants did not feel that they had to put considerable effort to accomplish the tasks. Low frustration levels indicate that the experiment design was not a significant barrier to accomplishing the task, further supported by the favorable responses regarding the virtual environment and the teleoperation of the UAV. At the same time, frustration was higher when prior knowledge about the missing person was not available, likely because participants felt they could now focus their search on a smaller region. As is evident from their search times, participants correctly perceived improved performance when they had prior knowledge of missing person.

Performance dependence on knowledge and swarm distribution

As expected having prior knowledge improved success rate and reduced times-to-find the missing person. Within successful trials, however, it was only the knowledge of missing person that affected the time-to-find and not the knowledge of the trails on the map. There are several reasons that could be attributed to this: first it is likely that it was much easier to reach a certain POA and then search along the boundary than to move back and forth on a trail; second, because the sites were not counterbalanced between conditions it is possible the effects of prior knowledge on teleoperator performance were diluted by the few terrain dissimilarities (for example, we observe that for the condition NN, participants tended to follow the river), although the random placement of the missing target within a terrain would mitigate such an effect; and finally, participants could not see the trails on the inset which meant they had to refer back to the printed briefing to obtain this information, and this would require more effort and time than remembering which POA to look around and referring to them on the inset. Not showing trails on the inset was a design choice that was selected to keep overhead maps realistic as trails can be considered temporary information often obtained from local experts, topographic maps, and low-altitude imagery [74].

Spatial distribution of the robot swarm within the experiments was found to improve performance despite the fact that the swarm location was not related to the missing person location. In the briefing for cases where the prior knowledge about a human was provided, we emphasized that the missing person may be somewhere between the middle and outer POA. Therefore, it is possible that placement of the clustered swarm nudged the participants to spend less time searching within the smaller POA where the missing person was never actually located. This was partially supported by our secondary analysis where we find that teleoperator trajectories when the swarms were clustered were more concentrated within the two inner POAs likely increasing the chances of finding the missing person. At the same time participants clearly didn't feel influenced by such swarms as they indicated in their survey responses. We note that placing the swarm cluster near the missing person defeats the purpose of our study where initially no one, human or autonomous robot, knows where the missing person is located.

Dependence and estimation of prior knowledge, cognitive load, and SA on movement

Participant movement was found to be affected by prior knowledge. The teleoperated UAV had a reduced turn rate when person knowledge was available (H1). It is possible that participants were more certain of where to search if they had missing person knowledge leading to fewer turns in their trajectories. Cognitive load was found to be weakly correlated with movement (H2). This weak correlation could be partly due to the inherent noisiness in calculating cognitive load using EEG, and a relatively short baseline of 5–7 s. At the same time, since we were interested in correlation with movement, cognitive load measured as difference between a constant baseline and test times should not be affected by the baseline which can be collected prior to the experiment [50, 75] or after an experiment [76]. In this context, although we treated α power as a measure of situational awareness, it can additionally serve as a measure of cognitive load [70].

Pupil diameter as an alternative way to measure cognitive load also did not register any correlations, likely because pupil diameter is also sensitive to incident light which may be varying depending on location in the virtual environment. On the other hand, saccade frequency was weakly correlated with turn rate at an observation window of 5, 10, and 20 s suggesting that it may be more robust measure of cognitive load than pupil diameter in such scenarios.

Situational awareness (SA) was also found to be weakly correlated with movement behavior. Although both dwell time calculated from eye tracking data and α power calculated using EEG data were found to correlate with fraction of time turning while still, α power showed stronger correlations.

Under high-stress conditions, as with driving on the road [77, 78], teleoperators may display exaggerated movement patterns (such as increased speed or sharper turns) thereby producing an amplified physiological response. Such a response could enhance the detectability of SA- and cognitive-load related cues. Although this investigation specifically examined Level 1 SA (environmental perception) in a controlled virtual setting, elevated stress levels may impair higher-order SA (environmental comprehension and

predictive capability), potentially making movement-based inferences more robust in real-world scenarios.

An observation window size of 15 s was found to generate highest accuracy for the LSTM network. Interestingly, this was also the observation window size for which SA achieved a high correlation with the turning while still movement feature, although the same was not true for cognitive load. It's worth noting that a 15-second observation was also reported in another study on ground-robot based search in a laboratory environment [27], and therefore it is possible that this time presents a reliable measurement window to infer the human teleoperator's prior knowledge and state.

Whereas the Grand Canyon National Park provides a large array of topographical features, it is possible that the movement inference proposed in this work may need to be updated when considering different terrains. For example, changes in turn rate and speed may be more often, and not only due to prior knowledge, when the terrain has many obstacles. This may entail capturing a baseline measurement involving exploration only to isolate dependencies on prior knowledge and situational awareness. In other words, while the values observed here may not directly transfer to a new terrain, the approach of observing movement behavior and using it to characterize teleoperator state could still be applied. With respect to reduced visibility, we expect an increase in false positives, which could entail adding additional movement features such as rate of change in altitude, to turn rate, speed, and freezing behavior.

A related question may arise with respect to whether the results of this study extend to other domains such as ground-based or aquatic robot platforms. In this context, recent work has shown that behavioral cues can indeed be used to infer operator knowledge when driving ground robots [27]. This coupled with existing work relating behavior to expertise in teleoperating a manipulator [79], and the effect of mental workload on driving behavior [80] suggests that prior knowledge may also manifest in operating ground robots. Nevertheless, features like terrain complexity, obstacles, and vegetation may amplify or suppress the feature correlations and require further investigation. With respect to aquatic systems, the relatively slow movements of underwater vehicles may entail observing movement features at a longer time scale than 15 s to identify stronger correlations with teleoperator state.

Finally, we highlight that the movement features such as speed, turn rate, and freezing that are interpreted in this work are likely instinctual, and therefore require no operator training. At the same time, teleoperators could be taught specific gesture-like movement patterns which robotic swarms could recognize as nonverbal cues and act accordingly.

Adaptive search with movement cues

Adaptive search strategies were found to have shorter times-to-find and have nearly the same success rate as the spiral search strategy, an established strategy [73] that typically requires all robots to localize themselves in order to maintain a configuration. Expectedly, because adaptive search strategies are informed by instantaneous prior knowledge and situational awareness these have a higher control consumption than random or spiral search.

Effect of including terrain knowledge Although adaptive strategies had high success rates, utilizing terrain knowledge was least effective with smaller swarms (AsPkSA versus AsPkTkSA). This is likely because smaller swarms were most effective in increasing the field of view when searching for a missing person near the trail, whereas our setup had the missing person located slightly off the trail. When SA was not included (AsPk versus AsPkTk) the difference in success rate is low suggesting that situational awareness likely had an interaction effect with the terrain knowledge.

Effect of including situational awareness Among the adaptive search strategies, strategy AsPkSA which uses person knowledge only regulated with SA has the lowest success rate compared to strategy AsPk which does not use SA estimate; when observing the search strategies that use terrain knowledge inference, we see no benefit of regulating using SA estimate in success rate. The same is true for time-to-find which suggests that SA estimate did provide a significant advantage. It is possible that the weak correlation with turning while still in experiments was not able to reliably extract teleoperator SA, which was generally low as seen in sample estimates. It is also possible that the effect of prior knowledge outweighs that of SA in performance.

Effect of swarm size In the scenario where a swarm of five autonomous robots were used, the adaptive search strategy depended on the information available. If no prior knowledge is available (NN), AsPkTk outperformed all other strategies with 10 and 15 robots when considering both success rate and time-to-find. With only success rate as the performance metric, spiral search with 5 robots was most effective followed closely by AsPk. For the case where missing person knowledge was available (YN and YY), search strategy AsPkTkSA had the best success rate and average time-to-find. For cases where only terrain knowledge was available (NY) either AsPkTk or AsPk was suitable. For a scenario with 15 robots, the time-to-find was reduced significantly and strategies AsPkTk and AsPk have identical success rates with relatively lower times-to-find.

Comparison of Spiral Search and Adaptive search methods The simulation results highlight that the choice between adaptive and spiral search strategies should be made on mission priorities. While spiral search achieves near-perfect success rates, adaptive search strategies took less time-to-find, up to 60% less when both prior knowledge available to teleoperator, and 20% less when no prior knowledge is available. This lower time-to-find can prove substantial in the rescue stage of a time-sensitive search-and-rescue mission. Nevertheless, investigating the unsuccessful trials in adaptive strategies reveal that continuing to search in this manner will not necessarily increase success rate and therefore a hybrid switching strategy where autonomous robots switch to an inward spiral after a certain time may prove to be the most successful strategy overall.

Comparison with recent multi-UAV search strategies

With respect to recent works on WiSAR using multiple robots, we note that similar improvements in times-to-find have been reported with search strategies that employ path planning on probability maps that are updated based on terrain topology [81] and missing person behavior and psychology [45]. Specifically, [81] reports a 70% improvement in time-to-find with respect to a lawn-mower search, and [45] report a 30% improvement over a deterministic strategy. By comparison, we note a 60% improvement with respect to spiral search which is equivalent to a deterministic strategy. Furthermore, because the approach proposed here is primarily human dependent (relies on observation and distance to human teleoperated UAV), it does not require localization capabilities assumed present in a robot that must follow a predetermined path.

Limitations

In our experiments, situational awareness is measured through neurophysiological measures and can only capture level one SA (paying attention to relevant aspects of the environment). While we favored such an approach over a SAGAT questionnaire deployed using a freeze-framing to avoid disruptions in search, it also prevented us from extracting higher levels of SA which could have shown a stronger correlation with movement cues.

Another limitation of our setup is related to uncertainties that may arise from internal and external conditions. For example, there may be uncertainties related to sensor noise that measures position and velocity of other robots as well as that of the human teleoperated UAV which is used to infer teleoperator state. These may be addressed by filtering approaches such as a Kalman filter for linear systems and a particle filter for nonlinear systems [82, 83]. External disturbances in the form of ambient wind may also affect how the autonomous swarm responds to the teleoperator state. Even though we set up the experiments in a GPS-aware environment, searching within dense vegetation may lead to loss of the GPS signal for extended periods of time, producing location bias. These can be addressed by updating the swarm control algorithm to ensure that the robotic swarm stays connected wirelessly [84] and one or more of the robots has GPS information. Finally, multi-agent systems considered here assume perfect communication and consensus, however real systems may suffer from dropouts and parametric uncertainties. Recent advances in iterative learning [85] and fuzzy control [86] address these types of uncertainties and could be integrated into this approach to make it more robust.

The adaptive search strategies presented here are limited to a one-way non verbal communication between teleoperator and robot swarm i.e. the swarm reacts to operator's movements but not the other way. In our experiments however, we find that the teleoperator performance does get affected by the presence of the swarm. A two-way non-verbal communication where as a swarm "reads" operator cues, it also changes its own spatial distribution for the operator to "read out" may offer improved performance in mixed-initiative strategies. We are investigating nonverbal communication from swarm to the teleoperator as part of ongoing work.

Conclusion and future directions

In this study we tested hypotheses related to whether movement of a teleoperated UAV can be used to estimate the teleoperator's prior knowledge, cognitive load and situational awareness, and whether such inference can be used to steer a swarm of autonomous robots. We found that prior knowledge, cognitive load, and situational awareness all correlate with teleoperated robot's movement in terms of speed, turn rate, and freezing behavior, and these can be used to infer the same during a search mission. Adaptive search strategies where an autonomous swarm inferred prior knowledge and situational awareness from robot movement had more success in finding the missing target than random search and lower times-to-find than spiral search.

While we selected Grand Canyon Park for its relevance in missing person search and varied topography, other movement features such as change in altitude for investigating possible sites may play a critical role in inferring operator state in low visibility situations and dense forest canopies.

There are two immediate extensions of this work that build on the results described here. First, based on our experimental results where a clustered swarm placed randomly within the search region was able to improve performance in terms of time-to-find, it is worth formulating and validating a two-way nonverbal communication strategy between the swarm and the teleoperator, where, despite being unaware of missing person location, a clustered swarm could nudge the teleop-

erator from searching in the wrong regions. Second, as with cognitive load and SA, it is likely that gaze direction (different from direction of motion of the UAV) may be estimated based on movement cues. For example, an experiment can be designed to identify where an operator may be looking when they are slowing down or freezing. Such inference can then serve as a cue to the robot swarm to proactively search in areas of interest within the region. Finally, there are several fault-detection algorithms that identify and isolate faults on the basis of movements or oscillations within robotic systems [87–89]. These same algorithms can be integrated within a robotic swarm to assist or take over from a human teleoperated robot when the onset of a fault is detected.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s40747-025-02055-9.

Acknowledgements The authors would like to thank Abdullah Thahab for authoring the grand canyon virtual environment. This research was partially supported by National Science Foundation under Grant # IIS-2033918.

Author Contributions Arunim Bhattacharya: Methodology, Software, Validation, Formal analysis, Writing - Original Draft. Sachit Butail: Conceptualization, Methodology, Formal analysis, Writing - Original Draft, Writing - Review & Editing, Supervision, Project administration, Funding acquisition.

Funding This research was supported by National Science Foundation under Grant # IIS-2033918.

Data availability All data generated or analysed during this study are included in this published article (and its supplementary information files).

Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethics approval The study was approved by the Institutional Review Board at Northern Illinois University under protocol # HS23-0233.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication The authors affirm that human research participants provided informed consent for publication of the videos recorded during the experiments.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

References

1. Heggie TW, Amundson ME (2009) Dead men walking: search and rescue in us national parks. Wilderness Environ Med 20(3):244–249. https://doi.org/10.1580/08-weme-or-299r.1
2. Doherty PJ, Guo Q, Doke J, Ferguson D (2014) An analysis of probability of area techniques for missing persons in Yosemite National Park. Appl Geogr 47:99–110. https://doi.org/10.1016/j.apgeog.2013.11.001
3. Dick A (2023) Arizona basic search and rescue. http://www.saraz.org/documents/AZBasicSAR.pdf
4. Tuśnio N, Wróblewski W (2021) The efficiency of drones usage for safety and rescue operations in an open area: a case from Poland. Sustainability 14(1):327. https://doi.org/10.3390/su14010327
5. Goodrich MA, Morse BS, Gerhardt D, Cooper JL, Quigley M, Adams JA, Humphrey C (2007) Supporting wilderness search and rescue using a camera-equipped mini UAV. J Field Robot 25(1–2):89–110. https://doi.org/10.1002/rob.20226
6. Grogan S, Pellerin R, Gamache M (2018) The use of unmanned aerial vehicles and drones in search and rescue operations—a survey. In: Proceedings of the PROLOG, pp 1–13
7. Silvagni M, Tonoli A, Zenerino E, Chiaberge M (2016) Multipurpose UAV for search and rescue operations in mountain avalanche events. Geomat Nat Haz Risk 8(1):18–33. https://doi.org/10.1080/19475705.2016.1238852
8. Nunnally S, Walker P, Kolling A, Chakraborty N, Lewis M, Sycara K, Goodrich M (2012) Human influence of robotic swarms with bandwidth and localization issues. In: IEEE international conference on systems, man, and cybernetics (SMC). https://doi.org/10.1109/icsmc.2012.6377723
9. Delmerico J, Mintchev S, Giusti A, Gromov B, Melo K, Horvat T, Cadena C, Hutter M, Ijspeert A, Floreano D, Gambardella LM, Siegwart R, Scaramuzza D (2019) The current state and future outlook of rescue robotics. J Field Robot 36(7):1171–1191. https://doi.org/10.1002/rob.21887
10. Isaacs J, Knoedler K, Herdering A, Beylik M, Quintero H (2022) Teleoperation for urban search and rescue applications. Field Robot 2(1):1177–1190. https://doi.org/10.55417/fr.2022039
11. Niedzielski T, Jurecka M, Miziński B, Remisz J, Ślopek J, Spallek W, Witek-Kasprzak M, Kasprzak Ł, Świerczyńska-Chlaściak M (2018) A real-time field experiment on search and rescue operations assisted by unmanned aerial vehicles. J Field Robot 35(6):906–920. https://doi.org/10.1002/rob.21784
12. Koester RJ (2008) Lost person behavior: a search and rescue guide on where to look for land, air, and water. DBS Productions, Charlottesville
13. Burke JL, Murphy RR, Coovert MD, Riddle DL (2004) Moonlight in Miami: field study of human–robot interaction in the context of an urban search and rescue disaster response training exercise. Hum-Comput Interact 19(1–2):85–116. https://doi.org/10.1080/07370024.2004.9667341
14. Syrotuck WG (2000) Analysis of lost person behavior. Barkleigh Productions, Mechanicsburg
15. Phillips K, Longden MJ, Vandergraff B, Smith WR, Weber DC, McIntosh SE, Wheeler AR III (2014) Wilderness search strategy and tactics. Wilderness Environ Med 25(2):166–176. https://doi.org/10.1016/j.wem.2014.02.006

16. Lin L, Goodrich MA (2010) A Bayesian approach to modeling lost person behaviors based on terrain features in wilderness search and rescue. Comput Math Organ Theory 16(3):300–323. https://doi.org/10.1007/s10588-010-9066-2
17. Hashimoto A, Heintzman L, Koester R, Abaid N (2022) An agent-based model reveals lost person behavior based on data from wilderness search and rescue. Sci Rep. https://doi.org/10.1038/s41598-022-09502-4
18. Kolling A, Walker P, Chakraborty N, Sycara K, Lewis M (2016) Human interaction with robot swarms: a survey. IEEE Trans Hum-Mach Syst 46(1):9–26. https://doi.org/10.1109/THMS.2015.2480801
19. Krzysiak R, Butail S (2021) Information-based control of robots in search-and-rescue missions with human prior knowledge. IEEE Trans Hum-Mach Syst 52(1):52–63. https://doi.org/10.1109/THMS.2021.3113642
20. Heintzman L, Hashimoto A, Abaid N, Williams RK (2021) Anticipatory planning and dynamic lost person models for human–robot search and rescue. In: IEEE international conference on robotics and automation (ICRA), pp 8252–8258. https://doi.org/10.1109/ICRA48506.2021.9562070
21. Jones B, Tang A, Neustaedter C (2020) Remote communication in wilderness search and rescue: Implications for the design of emergency distributed-collaboration tools for network-sparse environments. In: Proc ACM hum.-comput. interact., vol 4 (GROUP). https://doi.org/10.1145/3375190
22. Rasmussen ND, Morse BS, Goodrich MA, Eggett D (2009) Fused visible and infrared video for use in wilderness search and rescue. In: Workshop on applications of computer vision (WACV), pp 1–8. https://doi.org/10.1109/WACV.2009.5403048
23. Yanco HA, Drury J (2004) "Where am I?" Acquiring situation awareness using a remote robot platform. In: IEEE international conference on systems, man and cybernetics (IEEE Cat. No.04CH37583), vol 3, pp 2835–2840. https://doi.org/10.1109/ICSMC.2004.1400762
24. Sorensen L, Stanton N (2013) Y is best: How distributed situational awareness is mediated by organisational structure and correlated with task success. Saf Sci 56:72–79. https://doi.org/10.1016/j.ssci.2012.05.026
25. Endsley MR (2021) A systematic review and meta-analysis of direct objective measures of situation awareness: a comparison of SAGAT and SPAM. Hum Factors J Hum Factors Ergon Soc 63(1):124–150. https://doi.org/10.1177/0018720819875376
26. Nagi J, Giusti A, Gambardella LM, Di Caro GA (2014) Human-swarm interaction using spatial gestures. In: 2014 IEEE/RSJ international conference on intelligent robots and systems. IEEE, pp 3834–3841. https://doi.org/10.1109/IROS.2014.6943101
27. Krzysiak R, Butail S (2024) Human prior knowledge estimation from movement cues for information-based control of mobile robots during search. ACM Trans Hum Robot Interact. https://doi.org/10.1145/3706065
28. Ashour R, Aldhaheri S, Abu-Kheil Y (2023) Applications of UAVs in search and rescue. Springer, Cham, pp 169–200. https://doi.org/10.1007/978-3-031-32037-8_5
29. Shi Y, Kang J, Xia P, Tyagi O, Mehta RK, Du J (2021) Spatial knowledge and firefighters' wayfinding performance: a virtual reality search and rescue experiment. Saf Sci 139:105231. https://doi.org/10.1016/j.ssci.2021.105231
30. Clark DG, Ford JD, Tabish T (2018) What role can unmanned aerial vehicles play in emergency response in the arctic: a case study from Canada. PLoS ONE 13(12):0205299. https://doi.org/10.1371/journal.pone.0205299
31. Nourbakhsh IR, Sycara K, Koes M, Yong M, Lewis M, Burion S (2005) Human–robot teaming for search and rescue. IEEE Pervasive Comput 4(1):72–78. https://doi.org/10.1109/mprv.2005.13
32. Goodrich MA, Cooper JL, Adams JA, Humphrey C, Zeeman R, Buss BG (2007) Using a mini-UAV to support wilderness search and rescue: practices for human–robot teaming. In: International workshop on safety, security and rescue robotics, pp 1–6. https://doi.org/10.1109/SSRR.2007.4381284
33. Heintzman L, Williams RK (2021) A predictive autonomous decision aid for calibrating human-autonomy reliance in multi-agent task assignment. https://doi.org/10.48550/arXiv.2112.10252
34. Baker M, Casey R, Keyes B, Yanco HA (2004) Improved interfaces for human–robot interaction in urban search and rescue. In: International conference on systems, man and cybernetics (IEEE Cat. No.04CH37583), vol 3, pp 2960–29653. https://doi.org/10.1109/ICSMC.2004.1400783
35. Riley JM, Endsley MR (2004) The hunt for situation awareness: human–robot interaction in search and rescue. Proc Hum Factors Ergon Soc Annu Meet 48(3):693–697. https://doi.org/10.1177/154193120404800389
36. Wegner R, Anderson J (2006) Agent-based support for balancing teleoperation and autonomy in urban search and rescue. Int J Robot Autom 21(2):120–128. https://doi.org/10.2316/Journal.206.2006.2.206-2796
37. Hardin B, Goodrich MA (2009) On using mixed-initiative control: a perspective for managing large-scale robotic teams. In: Proceedings of the international conference on human robot interaction, pp 165–172. https://doi.org/10.1145/1514095.1514126
38. Guo Y, Jena R, Hughes D, Lewis M, Sycara K (2021) Transfer learning for human navigation and triage strategies prediction in a simulated urban search and rescue task. In: 30th IEEE international conference on robot & human interactive communication (RO-MAN), pp 784–791. https://doi.org/10.1109/RO-MAN50785.2021.9515526
39. Scerri P, Pynadath D, Tambe M (2001) Adjustable autonomy in real-world multi-agent environments. In: Proceedings of the fifth international conference on autonomous agents, pp 300–307. https://doi.org/10.1145/375735.376314
40. Scherer J, Yahyanejad S, Hayat S, Yanmaz E, Andre T, Khan A, Vukadinovic V, Bettstetter C, Hellwagner H, Rinner B (2015) An autonomous multi-UAV system for search and rescue. In: Proceedings of the workshop on micro aerial vehicle networks, systems, and applications for civilian use, pp 33–38. https://doi.org/10.1145/2750675.2750683
41. Hong A, Igharoro O, Liu Y, Niroui F, Nejat G, Benhabib B (2019) Investigating human–robot teams for learning-based semi-autonomous control in urban search and rescue environments. J Intell Robot Syst 94:669–686. https://doi.org/10.1007/s10846-018-0899-0
42. Ognibene D, Mirante L, Marchegiani L (2019) Proactive intention recognition for joint human–robot search and rescue missions through Monte-Carlo planning in POMDP environments. In: Social robotics. Springer, Cham, pp 332–343. https://doi.org/10.1007/978-3-030-35888-4_31
43. Britton BK, Tesser A (1982) Effects of prior knowledge on use of cognitive capacity in three complex cognitive tasks. J Verbal Learn Verbal Behav 21(4):421–436. https://doi.org/10.1016/s0022-5371(82)90709-5
44. Mohibullah W, Julie SJ (2013) Developing an agent model of a missing person in the wilderness. In: 2013 IEEE international conference on systems, man, and cybernetics. IEEE, Manchester. https://doi.org/10.1109/smc.2013.759
45. Macwan A, Vilela J, Nejat G, Benhabib B (2014) A multirobot path-planning strategy for autonomous wilderness search and rescue. IEEE Trans Cybern 45(9):1784–1797. https://doi.org/10.1109/TCYB.2014.2360368
46. Paas F, Tuovinen JE, Tabbers H, Van Gerven PWM (2003) Cognitive load measurement as a means to advance cognitive load theory. Educ Psychol 38(1):63–71

47. Colin TR, Smets NJJM, Mioch T, Neerincx MA (2014) Real time modeling of the cognitive load of an urban search and rescue robot operator. In: Proceedings of the international symposium on robot and human interactive communication. IEEE, Edinburgh. https://doi.org/10.1109/roman.2014.6926363
48. Moncur B, Galvez Trigo MJ, Mortara L (2023) Augmented reality to reduce cognitive load in operational decision-making. In: International conference on human-computer interaction. Springer, Cham, pp 328–346. https://doi.org/10.1007/978-3-031-35017-7_21
49. Sweller J, Ayres P, Kalyuga S (2011) Measuring cognitive load. Springer New York, New York, pp 71–85. https://doi.org/10.1007/978-1-4419-8126-4_6
50. Anderson EW, Potter KC, Matzen LE, Shepherd JF, Preston GA, Silva CT (2011) A user study of visualization effectiveness using EEG and cognitive load. Comput Graph Forum 30(3):791–800. https://doi.org/10.1111/j.1467-8659.2011.01928.x
51. Bales G, Kong Z (2018) Cognitive correlates of EEG spectral power indicate human-swarm task performance. In: Proceedings of the international conference on the Internet of Things. ACM, Santa Barbara. https://doi.org/10.1145/3277593.3277613
52. Mahanama B, Jayawardana Y, Rengarajan S, Jayawardena G, Chukoskie L, Snider J, Jayarathna S (2022) Eye movement and pupil measures: a review. Front Comput Sci 3:733531. https://doi.org/10.3389/fcomp.2021.733531
53. Bhattacharya A, Butail S (2023) Measurement and analysis of cognitive load associated with moving object classification in underwater environments. Int J Hum-Comput Interact 40(10):2725–2735. https://doi.org/10.1080/10447318.2023.2171275
54. Zagermann J, Pfeil U, Reiterer H (2018) Studying eye movements as a basis for measuring cognitive load. In: Extended abstracts of the CHI conference on human factors in computing systems, pp 1–6. https://doi.org/10.1145/3170427.3188628
55. Klingner J, Kumar R, Hanrahan P (2008) Measuring the task-evoked pupillary response with a remote eye tracker. In: Proceedings of the symposium on eye tracking research & applications. ACM, New York, pp 69–72. https://doi.org/10.1145/1344471.1344489
56. Krejtz K, Duchowski AT, Niedzielska A, Biele C, Krejtz I (2018) Eye tracking cognitive load using pupil diameter and microsaccades with fixed gaze. PLoS ONE 13(9):0203629. https://doi.org/10.1371/journal.pone.0203629
57. Endsley MR (1995) Toward a theory of situation awareness in dynamic systems. Hum Factors 37(1):32–64. https://doi.org/10.1518/001872095779049543
58. Lee JD (2008) Fifty years of driving safety research. Hum Factors 50(3):521–528. https://doi.org/10.1518/001872008X288376
59. Zhang T, Yang J, Liang N, Pitts BJ, Prakah-Asante K, Curry R, Duerstock B, Wachs JP, Yu D (2023) Physiological measurements of situation awareness: a systematic review. Hum Factors 65(5):737–758. https://doi.org/10.1177/0018720820969071
60. Paletta L, Dini A, Murko C, Yahyanejad S, Schwarz M, Lodron G, Ladstätter S, Paar G, Velik R (2017) Towards real-time probabilistic evaluation of situation awareness from human gaze in human–robot interaction. In: Proceedings of the international conference on human–robot interaction, pp 247–248. https://doi.org/10.1145/3029798.3038322
61. Hasanzadeh S, Esmaeili B, Dodd MD (2018) Examining the relationship between construction workers' visual attention and situation awareness under fall and tripping hazard conditions: using mobile eye tracking. J Constr Eng Manag 144(7):04018060. https://doi.org/10.1061/(ASCE)CO.1943-7862.0001516
62. Merchant S, Kwon Y, Schnell T, Etherington T, Vogl T (2001) Evaluation of synthetic vision information system (SVIS) displays based on pilot performance. In: Digital avionics systems conference, vol 1, pp 2–11. https://doi.org/10.1109/DASC.2001.963328
63. Zhou F, Yang XJ, De Winter JC (2021) Using eye-tracking data to predict situation awareness in real time during takeover transitions in conditionally automated driving. IEEE Trans Intell Transp Syst 23(3):2284–2295. https://doi.org/10.1109/TITS.2021.3069776
64. Lin C, He D, Kumar N, Choo K-KR, Vinel A, Huang X (2018) Security and privacy for the internet of drones: challenges and solutions. IEEE Commun Mag 56(1):64–69. https://doi.org/10.1109/MCOM.2017.1700390
65. Couzin ID, Krause J, James R, Ruxton GD, Franks NR (2002) Collective memory and spatial sorting in animal groups. J Theor Biol 218(1):1–11. https://doi.org/10.1006/jtbi.2002.3065
66. (2022) Cities-Skylines-heightmap-generator. https://github.com/sysoppl/Cities-Skylines-heightmap-generator
67. Papoulis A (1984) Probability, random variables and stochastic processes. McGraw Hill, Singapore
68. Codling EA, Plank MJ, Benhamou S (2008) Random walk models in biology. J R Soc Interface 5(25):813–834. https://doi.org/10.1098/rsif.2008.0014
69. Stone RT, Dorneich M, Gilbert S, McLaurin E (2013) Human differences in navigational approaches during tele-robotic search. Proc Hum Factors Ergon Soc Annu Meet 57:625–629. https://doi.org/10.1177/15419312135711
70. Klimesch W (1999) EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. Brain Res Rev 29(2–3):169–195. https://doi.org/10.1016/S0165-0173(98)00056-3
71. Salvucci DD, Goldberg JH (2000) Identifying fixations and saccades in eye-tracking protocols. In: Proceedings of the 2000 symposium on eye tracking research & applications, pp 71–78. https://doi.org/10.1145/355017.355028
72. Karim F, Majumdar S, Darabi H, Chen S (2018) LSTM fully convolutional networks for time series classification. IEEE Access 6:1662–1669. https://doi.org/10.1109/ACCESS.2017.2779939
73. Lyu M, Zhao Y, Huang C, Huang H (2023) Unmanned aerial vehicles for search and rescue: a survey. Remote Sens 15(13):3266. https://doi.org/10.3390/rs15133266
74. Bashyam A, Guggenheim J (2019) UAVs for wilderness search and rescue: Real-world considerations and technology roadmap for fixed wing UAVs. J Search Rescue 3(1):1–18. https://doi.org/10.61618/GLOX4896
75. Mahajan R, Majmudar CA, Khatun S, Morshed BI, Bidelman GM (2014) NeuroMonitor ambulatory EEG device: comparative analysis and its application for cognitive load assessment. In: 2014 IEEE healthcare innovation conference (HIC), pp 133–136. https://doi.org/10.1109/HIC.2014.7038892
76. Zarjam P, Epps J, Chen F (2010) Evaluation of working memory load using EEG signals. In: Proceedings of the APSIPA annual summit and conference, pp 715–719
77. Hill JD, Boyle LN (2007) Driver stress as influenced by driving maneuvers and roadway conditions. Transp Res F Traffic Psychol Behav 10(3):177–186. https://doi.org/10.1016/j.trf.2006.09.002
78. Yang L, Cheng J, Wang Z, Li X (2024) Assessing the relationship between driving skill, driving stress, and driving behavior. J Saf Res 91:96–107. https://doi.org/10.1016/j.jsr.2024.08.008
79. Enayati N, Ferrigno G, De Momi E (2018) Skill-based human–robot cooperation in tele-operated path tracking. Auton Robot 42:997–1009. https://doi.org/10.1007/s10514-017-9675-4
80. Paxion J, Galy E, Berthelon C (2014) Mental workload and driving. Front Psychol 5:1344. https://doi.org/10.3389/fpsyg.2014.01344
81. Giacomossi L, Maximo MR, Sundelius N, Funk P, Brancalion JF, Sohlberg R (2023) Cooperative search and rescue with drone swarm. In: Proceedings of the international congress and workshop on industrial AI. Springer, pp 381–393. https://doi.org/10.1007/978-3-031-39619-9_28
82. Bar-Shalom Y, Willett PK, Tian X (2011) Tracking and data fusion

83. Arulampalam MS, Maskell S, Gordon N, Clapp T (2002) A tutorial on particle filters for online nonlinear/non-Gaussian Bayesian tracking. IEEE Trans Signal Process 50(2):174–188. https://doi.org/10.1109/78.978374
84. Teacy WL, Nie J, McClean S, Parr G (2010) Maintaining connectivity in UAV swarm sensing. In: 2010 IEEE Globecom workshops. IEEE, pp 1771–1776. https://doi.org/10.1109/GLOCOMW.2010.5700246
85. Tao Y, Tao H, Zhuang Z, Stojanovic V, Paszke W (2024) Quantized iterative learning control of communication-constrained systems with encoding and decoding mechanism. Trans Inst Meas Control 46(10):1943–1954. https://doi.org/10.1177/01423312231225782
86. Du Z, Xie X, Qu Z, Hu Y, Stojanovic V (2024) Dynamic event-triggered consensus control for interval type-2 fuzzy multi-agent systems. IEEE Trans Circuits Syst I Regul Pap. https://doi.org/10.1109/TCSI.2024.3371492
87. Algburi RNA, Gao H, Al-Huda Z (2022) A new synergy of singular spectrum analysis with a conscious algorithm to detect faults in industrial robotics. Neural Comput Appl 34(10):7565–7580. https://doi.org/10.1007/s00521-021-06848-0
88. Sabry AH, Amirulddin UABU (2024) A review on fault detection and diagnosis of industrial robots and multi-axis machines. Results Eng 23:102397. https://doi.org/10.1016/j.rineng.2024.102397
89. Puchalski R, Giernacki W (2022) UAV fault detection methods, state-of-the-art. Drones 6(11):330. https://doi.org/10.3390/drones6110330

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.